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Invited Review in Celebration of the 50th Anniversary of EURO

Fifty years of multiple criteria decision analysis: From classical methods to robust ordinal regression

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ABSTRACT

Multiple Criteria Decision Analysis (MCDA) is a subfield of Operational Research that aims to support Decision-Makers (DMs) in the decision-making process through mathematical models and computational procedures. In this perspective, MCDA employs structured and traceable protocols to identify potential actions and the criteria for evaluating them. MCDA procedures aim to define recommendations consistent with the preferences of DMs for the specific decision problem at hand. These problems are generally formulated in terms of either choosing the best action, classifying actions into pre-defined and ordered decision classes, or ranking actions from best to worst. As the evaluation criteria are generally conflicting, the main challenge is to aggregate them into a mathematical preference model representing the DM value system. We review the development of MCDA over the past fifty years and describe its evolution with examples of distinctive methods. They are distinguished by the type of preference information elicited by DMs, the type of the preference model (criteria aggregation), and the way of converting the preference relation induced by the preference model in the set of potential actions into a decision recommendation. We focus on MCDA methods with a finite set of actions. References to specific application areas will be given. In the conclusion section, some prospective avenues of research will be outlined.

1. Introduction

Multiple Criteria Decision Analysis has been one of the fastest growing subfields of Operational Research since the 1970's. The European Journal of Operational Research (EJOR) has significantly contributed to this growth by publishing numerous impactful articles. For example, an early EJOR survey by Roy and Vincke (1981) outlined development prospects that were largely implemented in the following decades. One commonly divides the field of Multiple Criteria Decision Analysis into two subfields: 1) the case of a finite (known) set of potential actions (decision alternatives), and 2) multi-objective optimization, where the potential actions, formulated as vectors of decision variables, are implicitly defined using mathematical constraints (feasible solutions).

In the series of invited reviews published in EJOR to celebrate the 50th anniversary of the Association of European Operational Research Societies (EURO), there are two articles about Multiple Criteria Decision Analysis/Optimization, each focusing either on the problem of a finite set of potential actions, or the problem of multi-objective optimization. The division into deterministic problems and problems facing uncertainty are in principle relevant for both problems, but more common for the case of a finite set of potential actions. The multi-objective optimization literature seems to be focused on the deterministic case.

In this article, we review the developments of Multiple Criteria Decision Analysis over a period of fifty years, focusing on the case of a finite set of potential actions. We realize, however, that during this period, the two distinguished subfields influenced each other, particularly in the context of man–machine interaction and the learning of preferences. For example, both subfields commonly use pairwise (binary) comparisons of potential actions to acquire information about Decision-Maker (DM) preferences.

As the field of Multiple Criteria Decision Analysis has undergone a deep transformation during the half-century period, we will start our review with a description of "classical" methods for handling multiple criteria decision problems. These methods include the assessment of value/utility functions, the outranking approach, and the Analytic Hierarchy Process. Subsequently, we will move on to a new generation

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of methods that have emerged with the progress of computational technologies under the influence of the artificial intelligence paradigm, specifically learning preferences from data. This modern approach is referred to as Robust Ordinal Regression — it integrates many of the previous developments in the field and introduces new ways of building preference models, as well as new models themselves, such as decision rules.

We talk about decision making in general terms, and do not make a distinction between, for example, strategic and repetitive decisions. However, we feel that the artificial intelligence paradigm is particularly useful for repetitive decisions, since we can easily collect lots of preference data, and repeatedly use the same preference model, updated incrementally. For once-in-a-lifetime strategic decisions, one cannot fall back on previous decisions. Artificial intelligence can assist DMs even in such situations, but not by automating their decisions. Instead, it can aid them in understanding their preferences by analyzing some observed or simulated choices. Robust Ordinal Regression methods are examples of the latter situation.

Due to the conflicting nature of evaluation criteria, dealing with multiple criteria decision problems for the best choice, classification[1] or ranking requires information about the DM's preferences, which is subjective in nature. For this reason, the methodology of handling these problems is very different from the methodology of handling decision problems involving a single criterion to be optimized. While the latter is philosophically simple and objective, the former is mathematically "ill-posed" because the choice, classification or ranking is in this case dependent on the DM's preferences which are subjective, and there is no unique right way of including them in the aggregation of criteria. For this reason, Multiple Criteria Decision Analysis involving humans in the loop of interaction relies heavily on solution methodologies that are strongly dependent on culture and the language of communication. Unlike traditional Operational Research based on a single criterion to be optimized, Multiple Criteria Decision Analysis has created different schools and a plethora of approaches (Bragge et al., 2012). In very general terms, we can distinguish two main schools:

- The Multiattribute Utility/Value Theory School (Keeney & Raiffa, 1976) focused on a basic set of specifically understood rationality axioms. In particular, Multiattribute Utility Theory allows one to use the subjective expected utility model as the preference model, while Multiattribute Value Theory focuses on the deterministic case, by capturing DM's preferences via a multiattribute value function, which was often assumed to be additive. Many simpler approaches and heuristic procedures, such as SMART (Edwards, 1971), have subsequently been developed, seeking to be behaviorally more realistic. The biases discovered and investigated by the behavioral decision theory (Edwards, 1954; Kahneman & Tversky, 1984) are also considered and some specific procedures are proposed to correct them (Hammond et al., 1998). This school is often referred to as Anglo-Saxon, as it is typical of the culture of English-speaking countries with its characteristic pragmatism. The International Society on Multiple Criteria Decision Making[2] (MCDM), established in 1979, has adopted this perspective to a large extent, although its principal focus has been on optimization.
- The Multiple Criteria Decision Aiding (MCDA) School (Roy, 1996), based on the assumption that DMs' preferences do not pre-exist in their minds, and thus focused on decision support procedures aiming at co-constructing the preference model together with the DMs in a learning process. The goal of this process is not to discover an objectively best decision but to develop a corpus of convictions allowing the acceptance of a recommendation as the most suitable (Roy, 1993). This school, in turn, is often referred to as European (Roy & Vanderpooten, 1997), as it is typical of the culture of Continental European countries, especially France, with its characteristic emphasis on subtle distinctions. This perspective has been mainly adopted by the EURO Working Group on Multiple Criteria Decision Aiding,[3] established in 1975.

This distinction became increasingly blurred over time as the compelling arguments of each school spilled over into the other, and researchers from both schools collaborated and learned from each other at joint conferences and doctoral schools organized together. The developments of our field were presented in a history book by Köksalan et al. (2011), which covers both optimization and discrete, finite choice problems. On the other hand, the two volumes (Ehrgott et al., 2010; Greco, Ehrgott, & Figueira, 2016) presenting the state-of-the-art in our field, written by representatives of the two schools, witness the above-mentioned collaborations and convergence.

Our aim is twofold in this article: provide a representative overview of the evolution of Multiple Criteria Decision Analysis over the last fifty years and to present it in a tutorial manner for scholars who may not be intimately familiar with the field.

The rest of this article is organized as follows. In Section 2, we present a general view of Multiple Criteria Decision Analysis, including the building of the family of criteria serving to evaluate a finite set of potential actions, and preference handling using different preference models. Then, in Section 3, we present classical methods based on value/utility function preference models, and outranking methods using a system of binary relations to represent preferences. In Section 4, we overcome the assumption of preference independence between criteria, characteristic of the classical methods, and present methods using preference models handling interactions among criteria. In Section 5, the issue of preference information elicitation is presented, and in Section 6, the most recent model of preferences having the form of a set of "*if* ..., *then* ..." decision rules is presented. Section 7 characterizes the modern methods of Multiple Criteria Decision Analysis based on Robust Ordinal Regression. The final Section draws conclusions and outlines interesting avenues for future research.

2. General view of multiple criteria decision analysis

2.1. Decision analysis

Before we discuss Multiple Criteria Decision Analysis (MCDA), we wish to clarify how we understand Decision Analysis (DA). In the view of MCDA experts, DA aims to achieve the following general goals:

- Analyzing the context of the decision situation by identifying the problem in terms of its actors, potential actions, and their consequences, as well as the stakes. Among the actors, there are stakeholders (actors concerned by the decision), single or multiple decision-makers (users), and an analyst (facilitator). For the decision-maker we will use the abbreviation DM.
- Organizing and structuring how the decision-making process will unfold, to increase consistency between the values underlying the objectives, and the quality of the final decision.
- Generating recommendations based on results from models and computational procedures designed with respect to certain working hypotheses regarding DM preferences.
- Guiding the process to legitimize the final decision.

Decision problems considered by DA involve a finite set $\mathcal{A}$ of potential actions (alternatives, solutions, options, acts, candidates, etc.) evaluated from multiple points of view. The following three main problem statements are considered in DA:

[1] Also called sorting.
[2] https://www.mcdmsociety.org/
[3] https://www.cs.put.poznan.pl/ewgmcda/

- *Choice* of the best action.
- *Classification* (also called *sorting*) of actions into pre-defined and preference-ordered classes.
- *Ranking* of actions from best to worst.

With respect to classification problems, following Greco et al. (2001), Section 4, let us remark that this category of decision problems is divided into (i) problems where decision classes are not preference-ordered, called taxonomy problems, like plant taxonomy in botany, and (ii) problems where decision classes are preference-ordered, called ordinal classification, like classification of job candidates into accepted, invited for an interview, and rejected. Category (ii) can be further subdivided into (iia) where among the attributes describing the actions there are attributes with nominal scales, like classification of cars evaluated for their usefulness to a specific user, when cars have some non-ordinal features such as manufacturer or color, and (iib) where all attributes describing the actions have scales involving preference orders and are thus evaluation criteria. Category (iib) is called multiple criteria sorting, as reviewed, for example, by Zopounidis and Doumpos (2002). In the case of ordinal classification involving criteria, there is a semantic correlation between the criteria performance of an action and its class assignment. This monotonic relationship indicates that, with other performances held constant, better performance on a criterion should result in a class assignment that is not worse. For this reason the ordinal classification category involving criteria is often called ordinal classification with monotonicity constraints, particularly in the field of machine learning (Kotłowski & Słowiński, 2013). In the following, we will always refer to ordinal classification, using classification and sorting interchangeably. The above decision problems involve, in general, vector evaluations of actions, originating from:

- *Multiple decision-makers* (voters, group decision): decision-making process is often a multi-actor process.
- *Multiple evaluation criteria* (multiple objectives): a DM rarely has a single clear criterion in mind. Moreover, seldom is there a common unit of measurement for all criteria, which are rather heterogeneous. That is why it may be very difficult to define *a priori* a unique criterion able to take into account all relevant points of view.
- *Multiple possible states of the world* that imply multiple consequences of the actions. The problem setting is often non-deterministic and the outcome of a decision may then depend on uncertain scenarios. Probabilities for different states of nature may or may not be elicited.

The above origins of vector evaluations of actions gave birth, respectively, to three big subfields of decision theory:

- Theory of Social Choice (Arrow, 1951); for a recent presentation see (Fishburn, 2015).
- Multiple Criteria Decision Aiding (Roy, 1996) and Multiple Criteria Decision Making (Keeney & Raiffa, 1976); for state-of-the-art surveys see (Ehrgott et al., 2010; Greco, Ehrgott, & Figueira, 2016).
- Decision Making under Risk and Uncertainty (De Finetti, 1937; Ramsey, 1931; Von Neumann & Morgenstern, 1944); for a comprehensive survey see (Gilboa, 2009).

Although these sub-fields have become seemingly independent over the years, and developed their own tools of formal mathematical analysis, all start consideration of the above multidimensional decision problems from one common fact – the *dominance relation* in set $\mathcal{A}$ of actions: action a dominates action b if, on all dimensions considered, it is not worse than b, and there exists at least one dimension on which a is strictly better than b. This applies across all types of dimensions, i.e., all involved DMs in social choices, all criteria in MCDA problems, and all states of nature in decisions under risk and uncertainty. This is the only objective information that stems from the above problem statements. In case of decision-making under risk and uncertainty the concept is *stochastic dominance*: a stochastically dominates b if, for all utility levels, the probability of obtaining at least that level is not smaller with a than with b, and there exists at least one utility level where the probability of reaching that level is strictly greater with a. It is clear that the dominance relation is a partial preorder, i.e., a binary relation that is reflexive and transitive, but not necessarily complete. As such, the dominance relation leaves many actions incomparable, particularly when evaluations on multiple dimensions are in conflict, i.e., when improvement in one dimension causes deterioration in other dimensions. Incomparability prevents the formulation of an unambiguous recommendation in the best choice, classification or ranking. This is why the main preoccupation of these methodologies consists in aggregating the multiple dimensions into a *preference model* making the actions more comparable in light of users preferences.

2.2. The three-step methodology of decision analysis

All three sub-fields of the DA methodology try to "enrich" the dominance relation using preference information elicited from the DMs. Thus, the steps of the DA methodology that go beyond this point are the following:

- Acquire preference information from DMs exposing their value system by confronting them with the multidimensional evaluations of actions from set $\mathcal{A}$.
- Learn or construct a preference model that aggregates the vector evaluations of actions. The preference model generates a preference ordering in set $\mathcal{A}$, enabling a comparison of a greater number of actions than the dominance relation does.
- Transform the preference relation in set $\mathcal{A}$ into a recommendation of the best choice, classification, or ranking.

This three-step methodology may form loops in the case of *interactive methods*. In this article, we review various approaches to implementing the above three-step methodology in the field of Multiple Criteria Decision Analysis.

2.3. Building the family of criteria

In Multiple Criteria Decision Analysis (MCDA), the potential actions from set $\mathcal{A}$ are evaluated from multiple points of view, called *criteria* (or attributes), forming a family $\mathcal{G} = \{g_1, \dots, g_n\}$. Formally, the criterion is a real-valued function g_i defined on $\mathcal{A}$, reflecting the performance of each action from a particular point of view. This definition applies without loss of generality to qualitative criteria, where each ordered evaluation can be number-coded. In order to compare any two actions $a, b \in \mathcal{A}$ from this point of view, it is sufficient to compare two values: $g_i(a)$ and $g_i(b)$, called *performances*. It is worth stressing that criteria may have different evaluation scales: *ordinal*, when a difference of performance does not have the meaning of intensity, or *cardinal*, when a difference of performance has the meaning of intensity, for example of interval or ratio type. The type of criterion scale dictates the permissible arithmetic operations on the performances. For example, adding (weighted) performances on ordinal scales is not meaningful unless the ordinal evaluations are transformed into cardinal evaluations through a value function representing the DM's preferences. In the rest of this section, we assume, by default and without loss of generality, that criteria are of the *gain-type* (maximizing-type), i.e., the greater the performance, the better, unless explicitly specified that they are of the *cost-type* (minimizing-type).

Building the family of criteria $\mathcal{G}$ is the preliminary step of MCDA that precedes the elicitation of preferences. The role of family $\mathcal{G}$ is to:

- serve the purpose of communication and discussion during the decision process by structuring the evaluation of the decision problem,

- build the convictions and the feeling of satisfaction/dissatisfaction,
- guide the DMs towards a decision and contribute to rendering the decision legitimate.

Considered together, the n criteria of family $\mathcal{G}$ should satisfy certain desirable properties such as completeness, operationality, decomposability and non-redundancy proposed by Keeney and Raiffa (1976). Roy and Bouyssou (1993) proposed the following logical requirements of consistency:

- exhaustivity — if two actions have the same performances on all criteria, then they have to be indifferent, i.e., if for any $a, b \in \mathcal{A}$, $g_i(a) \sim_i g_i(b)$, $i = 1, \ldots, n$, then $a \sim b$,
- monotonicity — if action a is preferred to action b ($a \succ b$), and there is action c such that $g_i(c) \succsim_i g_i(a)$, $i = 1, \ldots, n$, then $c \succ b$,
- non-redundancy — elimination of any criterion from family $\mathcal{G}$ should violate at least one of the above properties,

where $\sim_i$, $\succ_i$ and $\succsim_i$ mean indifference, strict preference and weak preference with respect to $g_i \in \mathcal{G}$, while $\sim$, $\succ$ and $\succsim$ mean indifference, strict preference and weak preference in a comprehensive sense.

2.4. Handling preferences

Dealing with DM's preferences is at the core of MCDA. As mentioned in the Introduction, the development of MCDA over the past fifty years is marked by the existence of two schools approaching preference handling in different ways. These schools are known as Multiple Criteria Decision Aiding and Multiple Criteria Decision Making. Given that both schools operate within the same research domain, they often meet under the umbrella of Multiple Criteria Decision Analysis (Ehrgott et al., 2010).

Multiple Criteria Decision Making relies on normative and prescriptive Decision Analysis including tools for identifying, representing, and formally assessing important aspects of a decision, for prescribing a recommendation maximizing their utility (Bell et al., 1988). Decision Analysis assumes a kind of ideal rationality, and aims at giving an "objectively" best recommendation. It is based on 3 pillars:

- normative approach, which defines basic principles of rationality and deduces its consequences,
- descriptive approach, which verifies if these principles of rationality are respected in real-life decisions,
- prescriptive approach, which is concerned with helping DMs make better decisions; the emphasis is on usefulness of the approach; interestingly, as Keeney (1992) points out, the choice of the axioms is also a decision problem itself.

Multiple Criteria Decision Aiding starts from the observation that preferences are rarely clearly structured and known *a priori*. Hence, the questions of an analyst and the use of dedicated methods should be oriented towards shaping these preferences. In fact, this methodology proceeds by progressively forming a conviction and communicating about the foundations of these convictions. The models, procedures, and provided results constitute a communication and reflection tool. Indeed, they allow the participants of the decision process to carry forward their process of thinking, to discover what is important for them, and to learn about their values. Such elements of responses obtained by the DMs contribute to recommending and justifying a decision that increases the consistency between the evolution of the process and objectives and value systems of the stakeholders. Thus, Multiple Criteria Decision Aiding should be perceived as a *constructive learning process*, as opposed to machine discovery of an optimum of a pre-existing preference system or a truth that exists outside the involved stakeholders (Roy, 2005). This implies that the concepts, models, and methods proposed must not be considered as a means of discovering a pre-existing truth, which would be universally accepted. They have to be seen as keys to doors giving access to elements of knowledge contributing to the acceptance of a final recommendation (Roy, 1993).

The above says that Multiple Criteria Decision Aiding requires active participation of the stakeholders, which is often organized in an interactive process. In this process, phases of preference elicitation are interleaved with phases of computation of a recommended decision, in the same spirit of the early proposals of interactive multi-objective optimization methods, like Benayoun et al. (1971), Geoffrion et al. (1972) and Zionts and Wallenius (1976).

Summing up, Multiple Criteria Decision Aiding tries to:

- act in a constructive and transparent way, in interaction with the DMs, so that they elicit preference information reflecting adequately their evolving preferences,
- ensure a faithful representation of preferences,
- handle "imperfect" preference information, i.e., partial, inconsistent, unstable, uncertain.

It is interesting to note that the methodology of MCDA has co-evolved with the advancement of computing technology over the past fifty years. The most affected part of this methodology has been just the modeling of preferences:

- till early 1980's: "model-centric", i.e., first a model is chosen and then the DM is asked to provide preference information in terms of the model parameters,
- since 1980's: more and more "human-centric" due to appearance of personal computers allowing human–computer interaction, very often in a "trial-and-error" process,
- in XXI century: "knowledge-driven" due to wealth of data on human choices, i.e., holistic preference information is known first and then the model is built explaining the past decisions and predicting the future ones.

The knowledge-driven trend makes the preference modeling in MCDA similar to Preference Learning (Fürnkranz & Hüllermeier, 2010) in the Machine Learning paradigm. The differences, commonalities and synergies between Preference Learning and Multiple Criteria Decision Analysis have been recently analyzed in the review article (Hüllermeier & Słowiński, 2024).

The problem of preference handling in MCDA is outlined in the first two steps of the three-step DA methodology described in Section 2.2. Below, we will first characterize preference models and then the methods of obtaining information about preferences used to build these models.

2.5. Preference models

Preference models are synthesizing the multiple criteria evaluations of actions into a coherent framework that reflects the DM's preferences. The resulting preference model is inducing a preference relation in set $\mathcal{A}$, richer than the dominance relation.

One can distinguish three families of preference models synthesizing the multiple criteria evaluations in specific ways:

- Multiple Attribute Value Theory (MAVT) (Keeney & Raiffa, 1976) employing a value function that assigns a real value to each action $a \in \mathcal{A}$, e.g., the weighted sum of performances $U(a) = \sum_{i=1}^{n} k_i \times g_i(a)$, or a more general additive function $U(a) = \sum_{i=1}^{n} u_i[g_i(a)]$, with u_i being marginal value functions, or non-additive integrals handling interactions among criteria, like Choquet integral for cardinal criteria, and Sugeno integral for ordinal criteria (Grabisch, 1996).
- Outranking methods (Roy, 2005) employing a system of binary relations $\{\sim, \succ^w, \succ^s\}$ that together constitute the outranking relation $S = \{\sim \cup \succ^w \cup \succ^s\}$, where $\sim$ means indifference, $\succ^w$ weak preference, and $\succ^s$ strong preference; relation aSb reads: "action a is at least as good as action b".

- Decision rule models (Greco et al., 2001) consisting of logical statements induced from data, relating conditions on particular criteria and a decision, such as, for example, "if student Y is at least good in Mathematics and at least fair in Literature, then Y is overall at least medium", or "if student Y_1 is preferred to student Y_2 in Physics and at least weakly preferred in Philosophy, then Y_1 is comprehensively at least as good as Y_2".

The above preference models represent DM's preferences in different ways. A comparative analysis of these models reveals that the decision rule model requires the least stringent axioms, indicating that the value function and outranking models can represent specific preferences if and only if the rule model can as well (Słowiński et al., 2002). This assertion is also supported by the analysis performed in (Greco et al., 2004).

The classification rules having the syntax exemplified above are called *dominance-based rules* because in the condition part they involve a simple dominance relation between a partial performance profile of an action and a vector of threshold values, usually inferred by induction from data (Greco, Matarazzo, & Słowiński, 2016). These rules are the only aggregation operators that:

- capture most complex interactions among criteria,
- have non-compensatory nature treating criteria conditions individually,
- do not convert ordinal evaluation scales into cardinal ones, because the dominance relation only uses order.

Moreover, the rules reveal the values guiding DM's decisions, presenting each rule as a clear scenario outlining the causal relationship between performance on selected criteria and a comprehensive judgment.

3. Classical methods

3.1. Multiple attribute value and utility theory

3.1.1. Value and utility functions

Keeney and Raiffa's monumental book *Decisions with Multiple Objectives: Preferences and Value Tradeoffs* (Keeney & Raiffa, 1976) develops and summarizes the state-of-the-art of the field until mid-1970's. The book makes a difference between a value function and a utility function. Until then scholars had used the two terms interchangeably. In their terminology, value function refers to the deterministic situation, when we know the consequences of potential actions. Utility function refers to decision-making under uncertainty, capturing DM's risk preferences. Keeney and Raiffa's book is the most comprehensive source of reference about assessing value and utility functions in the presence of multiple criteria. Several of the theoretical results were developed in the 1960's by others, notably Debreu (1959, 1960), and Fishburn (1964, 1970). The authors recognize the need for preference elicitation in such a context. The concept of dominance, although very useful, is not enough to identify the most preferred decision or rank order considered actions.

We discuss value functions and utility functions separately. The concept of a tradeoff is central to both. The essence is in the question: "How much achievement on criterion g_i is the DM willing to give up in order to improve achievement on criterion g_j by a fixed amount?" If there is uncertainty in the problem context, the issue concerning the tradeoffs remains, but difficulties are magnified.

The value function is used to choose the most preferred action (providing highest value) or rank order the actions. The utility function is used to calculate the expected utility of each action. The expected utilities can be used to choose the most preferred action or rank order the actions.

3.1.2. Structuring of preferences via value functions

The usual technique to assess value functions is to break down the assessments into component parts. Much of the discussion and theory development focuses on value functions which can be expressed in additive form:

$$U(x_1, \dots, x_n) = \sum_{i=1}^{n} u_i(x_i),$$

where the u_i are marginal (single-dimensional) value functions, and $x_1, \dots, x_n$ are performances on criteria $g_1, \dots, g_n$. In order the additive form to hold, mutual preferential independence between any subset of criteria and its complement must hold (Debreu, 1960). Mutual preference independence means that the preference between any pair of actions, differing only in certain criteria, does not depend on fixed equal performances in the remaining criteria, i.e., more formally, that for all actions $a, b, c, d \in \mathcal{A}$ and $\mathcal{G}_1 \subset \mathcal{G}$, if $a \succsim b$, $g_i(a) = g_i(c)$ and $g_i(b) = g_i(d)$ for all $g_i \in \mathcal{G}_1$, $g_i(a) = g_i(b)$ and $g_i(c) = g_i(d)$ for all $g_i \in \mathcal{G} \backslash \mathcal{G}_1$, then $c \succsim d$. An alternative, and sometimes a more convenient form of an additive value function scales U, and u_i from 0 to 1, and introduces λ_i multipliers to the above summation, that is

$$U(x_1, \dots, x_n) = \sum_{i=1}^{n} \lambda_i \times u_i(x_i).$$

The additivity assumption can be weakened, but the assessment task becomes more complex.

A commonly used technique for assessing marginal value functions is to use the mid-value splitting technique, which is an alternative conjoint scaling procedure (Luce & Tukey, 1964). First, scale the value function so that $u_i(max) = 1$, and $u_i(min) = 0$. Then, identify the mid-value point on the x-axis (criterion scale) which provides $u_i(mid\text{-}point) = 0.5$. Then continue by splitting both portions to new mid-value points, that is, which correspond to 0.25 and 0.75 in terms of the value scale, and so forth, until we have a reasonably accurate single dimensional value function. The scaling constants λ_i can be estimated as follows. Ask the DM to compare vectors of criteria values and generate inequalities on the scaling constants. Or alternatively, one could construct two equally preferred degenerate lotteries, and from them generate a system of equations for the scaling constants being the unknowns.

Practicing decision scientists frequently use the so-called "pricing out" (or costing out) technique. This technique also assumes preferential independence. Moreover, this technique assumes that the marginal rate of substitution between money and any other criterion does not functionally depend on the monetary value. Note that one of the criteria (a reference criterion) is labeled as "money". Given a starting point (in terms of values of all criteria), the question is, how much would you be willing to pay (sacrifice, in terms of the reference criterion) to gain a unit improvement on another criterion.

3.1.3. Multiple attribute preferences under uncertainty

Different sets of axioms have been presented implying the existence of utilities, legitimizing expected utility as a good guide for making consistent decisions under uncertainty (see Fishburn, 1970; Savage, 1954; Von Neumann & Morgenstern, 1944, among others). The beauty of such utility functions is that they incorporate DM's attitude towards risk.

The Von Neumann method of standard gambles is the "standard" approach for assessing marginal (single-dimensional) utility functions. First, we scale the marginal utility function u from 0 to 1. Then, we present the DM with a set of gambles in the following format: what probability p would make you indifferent between receiving x amount for certain and a lottery, where you receive, say 100,000 dollars (max on the utility scale) with probability p and nothing with probability $(1 - p)$. Then, the indifference probability p is simply $u(x)$. Then, we repeat the exercise for a number of other x-values (typically 5–7). One can proceed also proposing the DM a 50%–50% lottery and asking for the equivalent amount x. For example, the DM can be asked "What

is the sure payoff x for which you would be indifferent between a lottery that gives a 50%–50% chance to 100,000 and 0 and the sure payoff x?", and, then $u(x) = 0.5 \times u(100,000) + 0.5 \times u(0)$. Note that in the lottery comparisons, the branches must be chosen so that the utilities of the corresponding consequences are known. A very reasonable (and common) assumption is the monotonicity of the marginal utility functions.

The utility function for money can be used to calculate many interesting measures related to risk aversion and risk-proneness (Pratt, 1978).

The more involved task is to choose an appropriate form for the multiple attribute (multiple criteria) utility function and elicit its parameters. Keeney and Raiffa (1976) have separate discussions for the two-attribute and multi-attribute cases.

The simplest form is the additive utility function:

$$U(x_1, \dots, x_n) = k_1 \times u_1(x_1) + \dots + k_n \times u_n(x_n), \quad k_i > 0, \quad i = 1, \dots, n, \quad \sum_{i=1}^{n} k_i = 1$$

where k_i are weights associated with u_i (marginal utility functions) which are normally scaled from 0 to 1. Utility functions are additive, if the following two conditions hold: (1) the criteria are mutually preferentially independent (see Section 3.1.2), and (2) the criteria are additively independent, i.e., the preference between actions depends only on the marginal probability distribution of performances on considered criteria. If we relax the strict additivity assumptions, but assume mutual utility independence, we end up with a more complex function, which includes product terms (the multiplicative utility function):

$$U(x_1, \dots, x_n) = \left(\prod_{i=1}^{n} \left(1 + k \times k_i \times u_i(x_i)\right) - 1 \right) \frac{1}{k},$$

with $k > -1$ and $k \neq 0$, being an additional scaling constant such that $1 + k = \prod_{i=1}^{n} (1 + k \times k_i)$.

Once we have assessed the marginal utility functions following the standard gambles method of Von Neumann, the next operational question is, how to decide which form is appropriate in your context. If preferences over lotteries on the criteria depend only on their marginal probability distributions for these criteria, and not on their joint probability distribution, then it is appropriate to assume the additive form. If the appropriate mutual utility independence assumptions hold, we can assume the multiplicative form. Keeney and Raiffa (1976) discuss how to operationalize utility independence and preferential independence tests.

The next question is, how to assess the scaling constants of both the additive function and the multiplicative function. The case of the additive function is relatively simple, and we outline it here. Criteria weights for the additive function are elicited by constructing two equally preferred degenerate lotteries: then ask the DM to define x_i such that $(x_1^0, \dots, x_i, x_{i+1}^0, \dots, x_n^0)$ is equally preferred to $(x_1^0, \dots, x_i^0, x_{i+1}^*, \dots, x_n^0)$, where superscript 0 implies the worst value of the criterion, and $*$ implies the best value (both scaled to be 0 and 1). Note that in the above lottery comparison all criteria values except for one are at the worst level. The preference statement implies that:

$$U(x_1^0, \dots, x_i, x_{i+1}^0, \dots, x_n^0) = U(x_1^0, \dots, x_i^0, x_{i+1}^*, \dots, x_n^0),$$

and $k_i \times u_i(x_i) = k_{i+1}$. In total, we need $n-1$ such comparisons plus the normalization constraint, to derive a unique set of weights.

The multiplicative case is doable, but is more complex, and we refer to Keeney and Raiffa (1976), pp. 347–348, for details. What if the underlying assumptions for either case do not hold? The advice from literature seems to be that, if they are not "too much" violated, we may still work with the additive or multiplicative form.

Value/utility theory has been the subject of active research for some 60 years. Besides Keeney and Raiffa's work, there are many other important contributions, such as Dyer and Sarin's measurable multiattribute value functions (Dyer & Sarin, 1979). Measurable value functions are based on the concept of "preference difference" between actions and provide an interval scale of measurement for preferences under certainty. They present conditions for additive, multiplicative, and more complex forms of the measurable multiple attribute value function, providing a link between the additive value function and multiple attribute utility theory.

In the 1950's and early 1960's Ward Edwards published two seminal articles that created behavioral decision research as a new field in psychology (Edwards, 1954, 1961). His focus was on how people make decisions and how these decisions can be improved with tools. In the first article (Edwards, 1954) introduced the expected utility model to psychologists and he asked: do people actually behave this way? Edwards' second paper (Edwards, 1961) with the title, "Behavioral Decision Theory", set the stage for a new field. Edwards' research in the seventies and eighties focused on developing useful decision analysis tools. He liked the Keeney–Raiffa's Multiattribute Utility Theory, but thought that it was not practical. Edwards' work included, among others, several versions of SMART, which he applied to social problems (Edwards, 1971, 1977).

The literature contains many illustrative and real applications of decision analysis based on value and utility functions. A classic example is the case of locating the new airport for Mexico City (De Neufville & Keeney, 1972).

3.2. The UTA method

Jacquet-Lagrèze and Siskos (1982, 2001) recognized the theoretical merits of Keeney and Raiffa's multiple attribute utility theory but thought that it lacked practical realism, which was also acknowledged in (Von Winterfelt, 1975). Keeping in mind some of the realistic properties of the MCDA school (Roy, 1977), they developed the UTA method (in French UTilité Additive). The UTA method does not assess a single value function, but a set of additive value functions. To do that, they propose the use of an ordinal regression method. It uses Linear Programming (LP) to optimally fit additive nonlinear value functions with preference information provided by the DM. As preference data Jacquet-Lagrèze and Siskos use ordinal preference comparisons (weak order) between so-called *reference actions*, and a subjective ranking of these actions. Typically, the set of reference actions is a subset of set $\mathcal{A}$, a set of past actions related to the same decision problem, a set of fictitious actions (performance vectors), or a combination of the above, and is familiar enough to the DM so that they are willing to express preferences for these actions.

The UTA method proceeds in two steps: (1) assessment of an "optimal" value function, which is compatible with the preference information, and (2) assessment of a set of value functions by means of post-optimality analysis. Step 2 is needed, because often the "optimal" utility function is not uniquely defined.

In step 1, the criteria scales are divided into equal intervals, making it possible to estimate the marginal value functions (assumed to be monotonically increasing) in a piecewise-linear form. The marginal value of an action is approximated by linear interpolation. Moreover, for each reference action two error terms are introduced, an overestimation error and an underestimation error of the action's additive value. The LP problem to identify the marginal value functions and weights k_i for the additive value function consists of an objective function (to be minimized), which is the summation of the total error in all preference statements, and a set of constraints consisting of the preference comparisons, requirement of monotonicity of the marginal value functions, and normalization constraints for the additive function and marginal value functions. Note that the estimation is done using piecewise-linear approximations of the marginal value functions.

The LP problem solved above does not necessarily lead to a unique solution. This is the case of the perfect fit (the total error minimized to 0), and the case of the imperfect approximation (the minimized total

error greater than 0). In both cases, many instances of value functions are either fully or partially consistent with the preference information. Due to the ambiguity of optimal solutions, Jacquet-Lagrèze and Siskos (1982, 2001) proposed the use of a stability (post-optimality) analysis to identify alternative optimal and suboptimal instances of the value function. They simply did it by adding one more constraint to the above LP problem, allowing the optimal objective function value (total error) to be "somewhat" higher than in the optimal solution. As result of this operation, they got a set of instances of the value function which rank the reference actions more or less similar to the DM (the similarity measure is Kendall distance between the ranking by the calculated value function and the ranking of the DM).

The two above papers contain references to some real-world applications. Since the original publication, the UTA method has been a subject of active research. It has also been extensively applied in practice. It is also worth mentioning the extension of the UTA methodology to ordinal classification problems (Jacquet-Lagrèze, 1995; Zopounidis & Doumpos, 1999).

To complete the discussion about the UTA method, it is interesting to compare it with the Zionts (1981) method, which is based on the continuous version of the Zionts and Wallenius (1976) method. Both methods address the same problem. The Zionts method uses pairwise comparisons of multicriteria actions and converges to the most preferred action. It also generates a set of feasible weights that satisfy the DM's preferences, assuming an additive linear value function. However, there are some differences in implementation: the UTA method operates with piecewise linear value functions. Although the goals appear similar, our interpretation is that the authors developed their respective approaches independently.

3.3. The analytic hierarchy process

The Analytic Hierarchy Process (for short AHP) was developed by Saaty in late 1970's (Saaty, 1980) to help DMs solve multiple criteria problems, where we know all potential actions evaluated by criteria having a hierarchical structure. It is one of the most popular decision-making approaches used by practitioners all over the world. Interestingly, Saaty's work at the State Department, and the Geneva negotiations with the Soviets concerning nuclear arms reduction served as a catalyst for Saaty to develop the AHP method (Saaty, 1968, 1980). The axiomatic foundation of AHP is treated in (Saaty, 1986).

The AHP decomposes the problem into a hierarchy of several smaller independent subproblems. On top of the hierarchy is the overall goal. The next layer consists of the main criteria decomposed into sub-criteria in the following layer, and so forth. The bottom layer consists of the actions.

Once the hierarchy has been constructed, the DM is expected to compare various elements (criteria, sub-criteria, and actions evaluated on each elementary criterion from the lowest level) to each other, two at a time, with respect to their impact on an element above them in the hierarchy.

The AHP asks the DM to make pairwise comparisons using the following subjective relative importance (ratio) scale: 1 = equal importance; 3 = moderate importance of one over the other; 5 = essential or strong importance; 7 = very strong importance; 9 = extreme importance; 2, 4, 6, and 8 represent intermediate values.

The DM is expected to compare each main criterion against each other. If there are sub-criteria, the DM is required to assess sub-criteria against each other (for each main criterion). Finally, the DM is asked to evaluate how well each action contributes towards each elementary criterion.

The information from the pairwise comparisons is synthesized by generating an importance weight for each element (criterion, action). To generate the weights, Saaty advocates the use of eigenvalues, although he recognizes many other ways to generate the weights. The eigenvalue formulation of the problem is $A\mathbf{w} = \lambda\mathbf{w}$, where matrix $A = [a_{ij}]$ represents the matrix of pairwise comparisons of importance of element i over j, $\mathbf{w} = [w_1, \dots, w_n]$ the vector of weights, and λ corresponds to the largest (or principal) eigenvalue.

Saaty also derives the Consistency Index:

$$CI = \frac{\lambda - n}{n - 1}.$$

Consistency index CI should be minimized and its optimal value is 0. It is reached when $\lambda = n$ and the pairwise comparison matrix A is perfectly consistent, i.e., when for all $i, j, k = 1, \dots, n$, $a_{ij} \times a_{jk} = a_{ik}$. Consistency index CI has to be compared with its expected value in case of randomly generated reciprocal matrices A of the same order n. The ratio between CI and the corresponding expected value is called consistency ratio CR, and if it is not less than 0.1, the advice is to study the problem and revise the judgments.

AHP has been criticized for a phenomenon known as rank reversal, where the preference order of actions may change when an action is added to or removed from the decision problem (Barzilai & Golany, 1994; Belton & Stewart, 2002). The method raised also a heated discussion for not respecting the condition of order preservation which requires that if $a_{ij} > a_{hk}$, then also $\frac{w_i}{w_j} > \frac{w_h}{w_k}$. Bana e Costa and Vansnick (2008) proved that the principal eigenvalue method used in AHP does not satisfy this condition.

Among the many extensions and generalizations of AHP, considering its widespread popularity, it is worth noting the Best–Worst Method (BWM) (Rezaei, 2015) which considers solely pairwise comparisons between each criterion and the best (most important) and worst (least important) criteria.

The Analytic Network Process (ANP) is a generalization of the AHP (Saaty, 1996). The basic structure is an influence network of clusters and nodes contained within the clusters. Priorities are established in the same way they are in the AHP using pairwise comparisons. A motivation for Saaty to develop ANP was that many decision problems cannot be structured hierarchically because they involve the interaction and dependence of higher-level elements in a hierarchy on lower-level elements. Not only does the importance of the criteria determine the importance of the actions in a hierarchy, but also the importance of the actions themselves determines the importance of the criteria. In the AHP, each element in the hierarchy is considered to be independent of all the others, that is, the criteria are considered to be independent of one another, and the actions are considered to be independent of the criteria and of each other. Saaty notes that in many real-world cases, there is interdependence among the items and the actions. ANP does not require independence among elements.

3.4. TOPSIS (Technique for Order of Preference by Similarity to Ideal Solution)

TOPSIS was originally developed by Hwang and Yoon (1981). It is based on the idea that the most preferred action should have the shortest geometric distance from an ideal action, and the longest geometric distance from an anti-ideal action. Let us remember that ideal action a^* and anti-ideal action a_* are fictitious actions having the best and the worst performances on all criteria, respectively:

$$g_i(a^*) = \max\{g_i(a), a \in \mathcal{A}\}, \quad g_i(a_*) = \min\{g_i(a), a \in \mathcal{A}\}, \; i = 1, \dots, n.$$

It uses a compensatory aggregation formula, where a poor result on one criterion can be compensated by a good result on another criterion. For each action $a \in \mathcal{A}$, the shorter distance $d(a, a^*)$ from a^* and the larger distance $d(a, a_*)$ from a_*, the more preferred is action a. The distances $d(a, a^*)$ and $d(a, a_*)$ are computed using normalized performances $\gamma_i(a)$, $a \in \mathcal{A}$, $i = 1, \dots, n$, obtained by dividing performance $g_i(a)$ by the square root of the sum of squared performances of actions from $\mathcal{A}$ on criterion g_i:

$$\gamma_i(a) = \frac{g_i(a)}{\sqrt{\sum_{a' \in \mathcal{A}} g_i(a')^2}}.$$

The distances $d(a, a^*)$ and $d(a, a_*)$, $a \in \mathcal{A}$, are of Euclidean type, and are calculated as follows:

$$d(a, a^*) = \sqrt{\sum_{i=1}^{n} \left(w_i \times \gamma_i(a) - w_i \times \gamma_1(a^*)\right)^2},$$

$$d(a, a_*) = \sqrt{\sum_{i=1}^{n} \left(w_i \times \gamma_i(a) - w_i \times \gamma_1(a_*)\right)^2}$$

where weights w_i, $i = 1, \ldots, n$, $w_i > 0$, are intended to differentiate the importance the DM wishes to assign to the various criteria in the aggregation formula. Using distances $d(a, a^*)$ and $d(a, a_*)$, $a \in \mathcal{A}$, one can calculate quality measure $v(a)$ for each action $a \in \mathcal{A}$ as follows:

$$v(a) = \frac{d(a, a_*)}{d(a, a^*) + d(a, a_*)}, \quad 0 \leqslant v(a) \leqslant 1$$

The actions from $\mathcal{A}$ can finally be ranked according to the decreasing value of quality measure $v(a)$, so that for $a', a'' \in \mathcal{A}$, a' is at least as good as a'' if and only if $v(a') \geqslant v(a'')$.

TOPSIS has found plenty of uses in industry. It seems easy to use, however, the criteria weights are not easy to understand as specific substitution rates in the Euclidean distance aggregation. TOPSIS does not provide a clear method by which to assign values for the weights. The weights are multiplied by normalized performances of the actions, and then the weighted performances appear in the Euclidean distance. Note that since the aggregation formula (weighted Euclidean distance) used in TOPSIS is different from the aggregation formula involving weights in AHP, one cannot apply AHP weights in TOPSIS, although this is a frequent methodological error by some authors. Indeed, the meaning of weights depends on the aggregation formula where they are used and, in general, they are not interchangeable between MCDA methods employing different aggregation formulas. Moreover, the use of the Euclidean (L_2) distance is somewhat arbitrary because, instead, any L_p distance could be used, giving, in general, a different result. TOPSIS has also been criticized for succumbing to the phenomenon known as rank reversal.

Martel and Roy (2006) made a thorough analysis of the meaningfulness of several aggregation formulas used in MCDA methods, including TOPSIS. They conclude: *To validate meaningful propositions using aggregation formulas used in TOPSIS, it is necessary for the performances to be measured on ratio scales. It is also necessary to have a very nuanced understanding of the complex meaning that these formulas confer to the weights w_i as well as to the characteristics of ideal and anti-ideal actions in order to be able to assign appropriate values to these parameters and interpret them correctly. Lastly, the fact that any modification of action performances must be accompanied by an adaptation of the weights w_i makes sensitivity analysis, and consequently, robustness analysis, nearly impractical.*

3.5. MACBETH

MACBETH was originally developed in the 1990's by Bana e Costa and Vansnick (1994, 1997). They adopt, in the UTA spirit, an additive value function model. It builds a quantitative model based on qualitative (verbal) judgments. In its spirit it is interactive and constructive (of preferences). MACBETH was motivated by the desire to avoid the aggregation difficulties inherent in ordinal aggregation (Condorcet paradox, Arrow's impossibility theorem).

MACBETH needs both ordinal information (x preferred over y) and information about the strength of preference concerning differences of attractiveness of actions being compared. To accomplish the latter, MACBETH uses the following questioning mode: When action x is more attractive than action y, "do you feel that the difference of attractiveness between x and y is "very weak", "weak", "moderate", "strong", "very strong", or "extreme".

MACBETH converts the verbal preference statements and the ordinal rankings into a quantitative scale (with values of actions placed on the scale) which is consistent with the preference statements. If the DM is not satisfied with the numerical values attached to the actions, they can interactively change them by moving a cursor. From the individual value scales, MACBETH builds an aggregate (additive) interval value scale, using weights represented by a histogram.

Many consulting projects have been performed using MACBETH. It is embedded in the M-MACBETH decision support software system.

3.6. VIMDA (Visual Interactive Method for Discrete Alternatives)

The approaches described above are suitable to DMs who need help solving problems consisting of a moderate number of actions and many criteria – sometimes possessing a hierarchical structure. However, they are less applicable to solving problems with a large (even huge) number (over a million) of actions, which are evaluated on a moderate number of criteria. VIMDA is a method and computer system developed by Korhonen (1988) for such large-scale problems.

The basic idea of VIMDA extends the Reference Point Method proposed by Wierzbicki (1980), which projects an aspiration point y (either feasible or infeasible) on the non-dominated set of actions using an achievement scalarizing function that for VIMDA is formulated as

$$f(a, y) = \max_{g_i \in \mathcal{G}} \left\{ w_i \times (y_i - g_i(a)) \right\},$$

where, for each criterion $g_i \in \mathcal{G}$, y_i is an aspiration level, and w_i is a scalarizing weight of criterion $g_i \in \mathcal{G}$, for example, $w_i = \frac{1}{max_{a \in \mathcal{A}}\{g_i(a)\}}$. A step-by-step description of the procedure follows:

1. Find a nondominated action in set $\mathcal{A}$ to start the search.
2. Ask the DM to tell which criterion values they want to improve, defining an aspiration point.
3. VIMDA generates a direction (called a reference direction) starting from the current action and passing through the virtual aspiration point.
4. The reference direction is projected onto the set of non-dominated actions using the achievement scalarizing function. This set (called a projected set) is provided to the DM for their evaluation using a graphical interface. Each point in this set is the "closest point" to the direction at a certain stage.
5. The DM is asked to choose the most preferred action from the projected set of actions. Name this action the current one. The search continues as long as the DM is willing to provide a new aspiration level for any of the criteria; otherwise, the procedure stops.

VIMDA is a "free search" method. It is not based on assuming a value function as a preference model, however, the minimized achievement scalarizing function assigns values to the projected set of actions and thus, together with the aspiration point, it represents the DM's preferences. The method operates by generating non-dominated actions for the DM's perusal in the region that suits their preferences. The DM directs the search by updating their preferences through the aspiration point. Each projection takes only a few seconds even for quite large data sets. Thus, the DM is able to make several iterations within a reasonable time limit. The search stops when the DM is satisfied with the current action. The entire interactive procedure is user-friendly and supported by a visual representation in computer graphics that allows the DM to evaluate a wide range of possible actions simultaneously.

3.7. Convex preference cones, and some extensions

The idea of convex preference cones is relatively simple (Nasrabadi et al., 2019). Assume we want to find the most preferred action from among a (large) set $\mathcal{A}$. Each action is evaluated with multiple criteria. The most preferred action depends on the DM's preferences. One approach is to use pairwise comparisons. There is some behavioral research, which indicates that such comparisons are relatively easy.

However, to identify the most preferred action with the help of pairwise comparisons may require a prohibitive number of such comparisons. We therefore searched for mechanisms, which would allow to eliminate many actions that, obviously, would not be of interest to the DM. Convex preference cones are one such mechanism.

Let us illustrate the method with an example. Let us assume that we simply randomly ask the DM to make a number of pairwise comparisons. Assume, for example, that action a is preferred to b, c is preferred to d, and e is preferred to d. Next, we group the actions into sets, where we know the worst action (in the set). This results in two sets: (1) a and b (b the worst), (2) c, d, and e (d the worst). Then, in the criterion space $X = \prod_{i=1}^{n} X_i \subseteq \mathbb{R}$, where X_i is the set of all possible evaluations on criterion $g_i \in \mathcal{G}$, we connect the action-points from the best to the worst (a to b; c to d and e to d), and extend the lines beyond the worst action-points. The spanning directions form the preference cones. In result, we get two preference cones: one two-point cone, and one three-point cone. As proved by Korhonen et al. (1984), all actions which lie in the cones or below them are not better than the known worst action - and can be eliminated from further consideration. The proof is based on assuming an increasing quasiconcave value function, but no other information about the value function is assumed.

In a nutshell, we generate a number of pairwise comparisons, and form as many preference cones (using the above logic) as we can. Then, LP is used to eliminate actions which lie in or under such convex preference cones.

Quasi-concave preference functions correspond to convex-to-the-origin indifference contours. They have been found to represent actual preferences well. The convex preference cone research was started by P. Korhonen, J. Wallenius, and S. Zionts in early 1980s but their first publication dealing with this topic emerged in 1984 (Korhonen et al., 1984). Over the years, these authors have collaborated with many colleagues on convex preference cones. Roughly a dozen publications are joint with M. Köksalan, who completed his Ph.D. on convex preference cones at State University of New York at Buffalo in 1983, with main results published in (Köksalan et al., 1984, 1986). Since then, Köksalan has worked with many Turkish Ph.D. students and colleagues on this topic, considering, for example, cones constructed on dummy actions to improve computational efficiency (Köksalan & Taner, 1992), or handling the case of a general monotone value function (Köksalan & Sagala, 1995). The use of preference cones in Evolutionary Multi-objective Optimization has been described in (Fowler et al., 2010). Convex preference cones can be used to generate a partial order or a total order (Dehnokhalaji et al., 2014, 2011). The collaboration with B. Lokman has resulted in two publications dealing with multiobjective integer programming problems, with the help of convex preference cones. An interesting paper reviewing the use of convex cones is (Karsu, 2011). Also see (Olson, 1996).

In a relatively recent paper, Karakaya et al. (2018) developed algorithms that are compatible with a broad family of preference functions. The algorithms are general in the sense that they can cover the whole family and can be made as restrictive as needed with the uncovered information regarding the parameters of the preference function. More specifically, they assume that underlying weighted L_α distance functions represent the DM's preferences. Such functions measure the distance between two actions using the weighted distance function. The authors use as an example the case where we wish to minimize the distance between the ideal action and a feasible action. L_α functions represent a special family of non-decreasing preference functions that can capture a wide variety of preference structures and can be considered as an approximation of quasiconvex family of preference functions. The authors use the DM's preferences in conjunction with the properties of these functions to eliminate the action space that cannot contain the most preferred action. The theory, depending on the structure of the DM's preferences, allows to reduce large portions of the action space and converge to the most preferred action quickly, in the spirit of convex preference cones.

3.8. Verbal decision analysis

Verbal Decision Analysis, developed by Oleg Larichev (Larichev & Moshkovich, 2013), is a methodological approach for aiding DMs with decisions involving multiple criteria. The main principles of the methodology are the use of natural language for problem description and the elicitation of preference information in ordinal form (e.g., "more preferable", "less preferable") to avoid quantitative estimations that are cognitively demanding for the DM and, consequently, unreliable. Several methods have been developed using the verbal decision approach, with the most well-known being ZAPROS (Larichev & Moshkovich, 1995) for ranking problems, and ORCLASS (Larichev & Moshkovich, 1994) for sorting problems.

3.9. Outranking methods

The concept of outranking relation (Roy, 1968, 1985, 1991) is based on the idea that it is reasonable to compare actions from $\mathcal{A}$ taking into account the possibility of incomparability, and situations when the DM hesitates between indifference and preference. This is not possible with methods that assign a single comprehensive value $V(a) = V(g_1(a), \ldots, g_n(a))$ to each action $a \in \mathcal{A}$. Indeed, in case of MAVT there is only space for a preference relation, when $V(a) > V(b)$, or indifference, when $V(a) = V(b)$. To permit representation of incomparabilities, hesitations and also intransitivities in comparison between actions from $\mathcal{A}$, outranking relations were introduced. In general, for $a, b \in \mathcal{A}$, aSb (a outranks b) if there are sufficiently strong arguments for considering a at least as good as b, and there is no sufficiently strong counterargument which would be able to invalidate this assertion. The outranking approach was introduced with the ELECTRE family of methods. Another class of very popular outranking methods is the PROMETHEE family. In addition to these families, other outranking methods have been proposed and recently reviewed by Martel and Matarazzo (2016). Below, we present both ELECTRE and PROMETHEE families of methods, but without details available in the source studies.

3.9.1. ELECTRE methods

ELECTRE methods have been developed by Bernard Roy and his collaborators since the mid-1960's. The history of ELECTRE methods is strictly related to many real world applications (for a detailed presentation see Roy and Bouyssou (1993); for state-of-the-art surveys see Figueira et al., 2013, 2016). In ELECTRE methods, the DM's preferences are modeled with different variants of outranking relation S, such that for all $a, b \in \mathcal{A}$, aSb means "*a is at least as good as b*". The outranking relation S is reflexive, but, in general, neither transitive nor complete. The outranking relation of ELECTRE methods can also be a fuzzy relation $S : \mathcal{A} \times \mathcal{A} \to [0, 1]$, such that for all $a, b \in \mathcal{A}$, $S(a, b)$ expresses the credibility of the statement "*a is at least as good as b*". To validate relation aSb, the following two conditions should be fulfilled:

1. *Concordance*: there is a *sufficiently* strong coalition of criteria for which a is at least as good as b.
2. *Non-discordance*: there is no criterion for which the advantage of b over a is so strong that it is not possible to say that a is comprehensively at least as good as b.

The above conditions are checked in the concordance test and the discordance test, respectively. In the ELECTRE methods, the construction of one or more outranking relations is followed by a procedure transforming the obtained outranking relations to a recommendation related to the specific decision problem dealt with, i.e., the best choice, classification or ranking.

Specific discrimination thresholds, called indifference threshold and preference threshold, are associated with evaluation criteria $g_i \in \mathcal{G}$ to consider the imperfect nature of evaluations $g_i(a)$ for actions $a \in \mathcal{A}$. Criteria equipped with indifference and preference thresholds are called

pseudocriteria (Roy & Vincke, 1984). When comparing the actions, they permit to distinguish indifference, weak preference and preference with respect to each criterion $g_i \in \mathcal{G}$. More precisely, for $a, b \in \mathcal{A}$ and $g_i \in \mathcal{G}$ with $g_i(a) \geq g_i(b)$:

1. Indifference threshold $q_i(g_i(b))$ is the greatest difference $g_i(a) - g_i(b)$ for which there is indifference between a and b with respect to criterion g_i.
2. Preference threshold $p_i(g_i(b))$ is the smallest difference $g_i(a) - g_i(b)$ for which a is preferred to b with respect to criterion g_i.
3. If $q_i(g_i(b)) < g_i(a) - g_i(b) < p_i(g_i(b))$, a is weakly preferred to b in the sense that there is hesitation between indifference and preference.

Considering $a, b \in \mathcal{A}$ and $g_i \in \mathcal{G}$, the following preference relations can be defined:

1. Action a is indifferent with action b with respect to g_i, denoted as aI_ib, if $-q_i(g_i(a)) \leqslant g_i(a) - g_i(b) \leqslant q_i(g_i(b))$.
2. Action a is preferred to action b with respect to g_i, denoted as aP_ib, if $g_i(a) - g_i(b) \geq p_i(g_i(b))$.
3. There is hesitation between indifference and preference of a over b with respect to g_i, denoted as aQ_ib, if $q_i(g_i(b)) < g_i(a) - g_i(b) < p_i(g_i(b))$.

Moreover, each criterion $g_i \in \mathcal{G}$ is assigned a weight w_i, $w_i \geq 0$, $\sum_{g_i \in \mathcal{G}} w_i = 1$, related to its "intrinsic" importance, reflecting the voting power of g_i in the coalition of criteria considered in the concordance test. In Appendix A, we present an operational method to determine the values of such weights (Figueira & Roy, 2002). Finally, in ELECTRE methods, there is another parameter underlying the importance of criterion g_i – it is called the veto threshold $v_i(g_i(a)) \geq p_i(g_i(a))$. This is the smallest difference $g_i(b) - g_i(a)$ in favor of action b permitting to invalidate the hypothesis that aSb, even if all $n-1$ criteria are supporting it. The veto thresholds are considered in the discordance test.

In the following we provide a synthetic description of the most representative ELECTRE methods: ELECTRE I, ELECTRE III and ELECTRE TRI, while other ELECTRE methods are briefly recalled at the end.

ELECTRE I There are several variants of this method: ELECTRE I, the first one and the simplest; ELECTRE Iv, that takes into consideration a veto threshold changing from one criterion to the other, and ELECTRE IS, the most general, taking into account imprecision in the evaluations $g_i(a)$ by means of pseudocriteria. Considering ELECTRE IS (Roy & Skalka, 1984), the concordance test is passed if

$$c(aSb) = \sum_{g_i \in \mathcal{G}:\ a(P_i \cup Q_i \cup I_i)b} w_i + \sum_{g_i \in \mathcal{G}:\ bQ_ia} \varphi_i \times w_i \geq s$$

where

$$\varphi_i = \frac{g_i(a) + p_i\big(g_i(a)\big) - g_i(b)}{p_i\big(g_i(a)\big) - q_i\big(g_i(a)\big)}$$

allows a gradual transition from full contribution to concordance (when $g_i(a) \geqslant g_i(b) - q_i(g_i(a))$) to absence of contribution (when $g_i(a) \leqslant g_i(b) - q_i(g_i(a))$), and s, such that $0.5 \leq s \leq 1 - min_{g_i \in \mathcal{G}}\{w_i\}$, is a concordance level that fixes the required strength of the coalition of criteria being in favor of the hypothesis aSb. The discordance test (no veto condition) is passed if for all $g_i \in \mathcal{G}$:

$$g_i(a) + v_i\big(g_i(a)\big) \geq g_i(b) + q_i\big(g_i(b)\big) \times \eta_i \tag{1}$$

where $v_i\big(g_i(a)\big)$ is the veto threshold and

$$\eta_i = \frac{1 - c(aSb) - w_i}{1 - s - w_i}$$

is a coefficient permitting to reinforce the veto effect according to the decreasing value of the concordance $c(aSb)$. In fact, basically, the no-veto condition should hold if

$$g_i(a) + v_i\big(g_i(a)\big) \geq g_i(b) \Leftrightarrow g_i(b) - g_i(a) \leq v_i\big(g_i(a)\big) \tag{2}$$

which is the formulation of this condition in the original ELECTRE I method (Roy, 1968). In ELECTRE IS, condition (1) permits to reject the outranking aSb even if condition (2) holds. In fact, condition (1) can be rewritten as

$$g_i(b) - g_i(a) \leq v_i\big(g_i(a)\big) - q_i\big(g_i(b)\big) \times \eta_i$$

so that to prevent the veto effect, it is not sufficient for the difference $g_i(b) - g_i(a)$ to be not greater than $v_i\big(g_i(a)\big)$ but rather not greater than $v_i\big(g_i(a)\big) - q_i\big(g_i(b)\big) \times \eta_i$.

In ELECTRE I, the procedure for transforming the outranking relation into a recommendation is based on the concept of a kernel, that is the subset of actions $\mathcal{K} \subseteq \mathcal{A}$ satisfying the following conditions:

1. Internal stability: all the actions in the kernel are incomparable, i.e., for all $a, b \in \mathcal{K}$ neither aSb nor bSa.
2. External stability: each action outside the kernel is directly outranked by at least one action in the kernel, i.e., for all $b \in \mathcal{A} \setminus \mathcal{K}$ there is at least one $a \in \mathcal{K}$ such that aSb.

In fact, the kernel exists and it is unique if and only if the outranking relation S does not have cycles. For this reason, several procedures were proposed to "reduce" the cycles. The kernel is finally presented to the DM as the set of best (incomparable) actions.

ELECTRE III ELECTRE III is using a fuzzy outranking relation S, such that, for all $a, b \in \mathcal{A}$, $S(a, b)$ can be interpreted as the credibility of the statement "a outranks b". $S(a, b)$ is defined taking into account the concordance index of ELECTRE IS and discordance index $d_i(a, b)$ for each criterion $g_i \in \mathcal{G}$, defined as follows:

$$d_i(aSb) = \begin{cases} 1 & \text{if} \quad g_i(b) > g_i(a) + v_i(g_i(a)) \\ 0 & \text{if} \quad g_i(b) \leq g_i(a) + p_i(g_i(a)) \\ \frac{g_i(b) - g_i(a) - p_i(g_i(a))}{v_i(g_i(a)) - p_i(g_i(a))}, & \text{otherwise.} \end{cases}$$

On the basis of these elements, ELECTRE III defines the fuzzy outranking relation $S(a, b)$ as follows:

$$S(a, b) = c(aSb) \prod_{\{g_i \in \mathcal{G}\ :\ d_i(aSb) > c(aSb)\}} \frac{1 - d_i(aSb)}{1 - c(aSb)}$$

Observe that $S(a, b) = c(aSb)$ if there is no criterion for which $g_i(b) > g_i(a) + p_i(g_i(a))$, and $S(a, b) = 0$ if there is one or more criteria for which $g_i(b) > g_i(a) + v_i(g_i(a))$. In all other cases $S(a, b)$ is obtained by multiplying $c(aSb)$ by a reduction coefficient $\frac{1 - d_i(aSb)}{1 - c(aSb)}$ for each criterion satisfying $d_i(aSb) > c(aSb)$.

ELECTRE III provides a recommendation in the form of a partial ranking of actions from $\mathcal{A}$ obtained as intersection of two complete rankings derived from two "distillation" procedures performed on the fuzzy outranking relation S (Roy, 1991). In particular, there is a descending distillation procedure, ordering the actions form best to worst, and an ascending distillation procedure, ordering the actions from worst to best. Pairs of actions that compare inversely in descending and ascending orders are incomparable in the final ranking.

ELECTRE TRI ELECTRE TRI permits to classify actions from $\mathcal{A}$ to a set of predefined preference-ordered classes $\mathcal{C} = \{C_1, \ldots, C_h, \ldots, C_k\}$, where C_{h+1} is a class better than C_h, $h = 1, \ldots, k-1$. Each action $a \in \mathcal{A}$ is assigned to a class from $\mathcal{C}$ by comparing a with reference profiles b_h, $h = 1, \ldots, k-1$, separating class C_h from class C_{h+1}. In particular, the fuzzy outranking relation S of ELETRE III is considered together with a *λ-cutting level* in order to obtain a crisp outranking relation for which a_1 is at least as good as a_2, a_1Sa_2, if $S(a_1, a_2) \geq \lambda$. The following two procedures have been proposed to make classification decisions:

- *Pessimistic rule*. Each action $a \in \mathcal{A}$ is assigned to the highest class C_h such that aSb_{h-1}, $h = 2, \ldots, k$; if for all $h = 1, \ldots, k-1$, not aSb_h, then a is assigned to C_1;

- *Optimistic rule*. Each action a is assigned to the lowest class C_h such that $b_h \succ a$, that is $b_h S a$ but not $a S b_h$; if for all $h = 1, \dots, k-1$, not $b_h \succ a$, then a is assigned to C_k.

Other ELECTRE methods ELECTRE II method provides a partial ranking of actions. It is based on two outranking relations from which two complete rankings Z_1 and Z_2 are obtained. Z_1 is iteratively ordering actions from $\mathcal{A}$ from best to worst, following the rule "no other is preferred to them", while Z_2 is iteratively ordering actions from worst to best, following the rule "they are not preferred to any other". The final ranking is obtained through intersection of Z_1 and Z_2.

ELECTRE IV provides also a partial ranking of actions from $\mathcal{A}$. Its main characteristic is that, acknowledging the difficulty of specifying the weights of criteria from $\mathcal{G}$, it constructs a set of five embedded outranking relations $S^1, \dots, S^5$ with a decreasing credibility, without assigning weights to criteria. In ELECTRE IV, the procedure for transforming the outranking relation into a partial ranking of actions is the same as in ELECTRE III.

ELECTRE TRI-C is a classification method taking into account central reference profiles instead of separating reference profiles used in ELECTRE TRI. The central reference profiles are (virtual) actions being characteristic for each class (Almeida-Dias et al., 2010).

ELECTRE TRI-NB (Fernández et al., 2017) and ELECTRE TRI-NC (Almeida-Dias et al., 2012) are two extensions of ELECTRE TRI and ELECTRE TRI-C methods, respectively. ELECTRE TRI-NB considers a plurality of reference profiles separating contiguous classes. ELECTRE TRI-NC considers a plurality of central reference profiles for each class.

ELECTRE SCORE (Figueira et al., 2022) assigns a score range to each action $a \in \mathcal{A}$ on the basis of a comparison of a with sets of reference actions.

3.9.2. PROMETHEE methods

This family of methods was developed starting from 1982 by Jean-Pierre Brans and collaborators (Brans & Vincke, 1985; Brans et al., 1986) (for an updated state-of-the-art survey on PROMETHEE methods see Brans & De Smet, 2016). PROMETHEE methods are based on multicriteria aggregated preference indices constructed as follows. For each criterion $g_i \in \mathcal{G}$, a preference function is defined:

$$P_i(a,b) = F_i[g_i(a) - g_i(b)], \quad a, b \in \mathcal{A}$$

with F_i being a specific non-decreasing function taking values in $[0,1]$ such that, with respect to criterion g_i,

- $P_i(a,b) = 0$ means that there is no preference between a and b,
- $P_i(a,b) = 1$ means that there is a clear and definite preference of a over b,
- in general, the greater $P_i(a,b)$, the greater the preference of a over b.

Function F_i has to be selected from the following six specific forms permitting to pass from indifference to preference with a desired continuity and smoothness, possibly taking into account indifference and preference non-negative constant thresholds, denoted by q_i and p_i, respectively:

- Type 1 (usual criterion): $P_i(a,b) = 0$ if $g_i(a) \le g_i(b)$ and $P_i(a,b) = 1$ otherwise.
- Type 2 (U-shape criterion): $P_i(a,b) = 0$ if $g_i(a) - g_i(b) \le q_i$ and $P_i(a,b) = 1$ otherwise.
- Type 3 (V-shape criterion): $P_i(a,b) = 0$ if $g_i(a) \le g_i(b)$, $P_i(a,b) = \frac{g_i(a)-g_i(b)}{q_i}$ if $0 < g_i(a) - g_i(b) \le q_i$, and $P_i(a,b) = 1$ otherwise;
- Type 4 (level criterion): $P_i(a,b) = 0$ if $g_i(a) - g_i(b) \le q_i$, $P_i(a,b) = 0.5$ if $q_i < g_i(a) - g_i(b) \le p_i$, and $P_i(a,b) = 1$ otherwise.
- Type 5 (V-shape with indifference criterion): $P_i(a,b) = 0$ if $g_i(a) - g_i(b) \le q_i$, $P_i(a,b) = \frac{g_i(a)-g_i(b)-q_i}{p_i-q_i}$ if $q_i < g_i(a) - g_i(b) \le p_i$, and $P_i(a,b) = 1$ otherwise.
- Type 6 (Gaussian criterion): $P_i(a,b) = 0$ if $g_i(a) - g_i(b) \le 0$ and $P_i(a,b) = 1 - e^{-\frac{(g_i(a)-g_i(b))^2}{2s_i^2}}$ otherwise, with s_i being an intermediate value between q_i and p_i such that the greater s_i the more softened the preference for small deviations.

On the basis of preferences $P_i(a,b)$, the aggregated preference index is defined

$$\pi(a,b) = \sum_{i=1}^{n} w_i \times P_i(a,b),$$

with $w_i \ge 0$ being the weight of relative importance of criterion $g_i \in \mathcal{G}$, $\sum_{i=1}^{n} w_i = 1$. $\pi(a,b)$ takes a value between 0, for which there is no preference of a over b, and 1, for which there is a strong and undoubtful preference of a over b. For all the intermediate values between 0 and 1, the greater $P(a,b)$, the greater the preference of a over b. Given set $\mathcal{A}$ containing m actions, taking into consideration the comparison of action $a \in \mathcal{A}$ with all other actions $b \in \mathcal{A} \setminus \{a\}$, the following outranking flows can be defined:

- the positive outranking flow, expressing the strength of a:

$$\phi^+(a) = \frac{1}{m-1} \sum_{b \in \mathcal{A} \setminus \{a\}} \pi(a,b), \tag{3}$$

- the negative outranking flow, expressing the weakness of a:

$$\phi^-(a) = \frac{1}{m-1} \sum_{b \in \mathcal{A} \setminus \{a\}} \pi(b,a), \tag{4}$$

- the net outranking flow, expressing the balance between the strength and the weakness of a:

$$\phi(a) = \phi^+(a) - \phi^-(a). \tag{5}$$

The PROMETHEE I method proposes the partial ranking of actions from $\mathcal{A}$ defined as follows for all $a, b \in \mathcal{A}$:

- a is comprehensively preferred to b if $\phi^+(a) \ge \phi^+(b)$ and $\phi^-(a) \le \phi^-(b)$, with at least one inequality being strict;
- a is indifferent with b if $\phi^+(a) = \phi^+(b)$ and $\phi^-(a) = \phi^-(b)$;
- a is incomparable with b, otherwise, that is,
  - if $\phi^+(a) > \phi^+(b)$ and $\phi^-(a) > \phi^-(b)$, or
  - if $\phi^+(a) < \phi^+(b)$ and $\phi^-(a) < \phi^-(b)$.

The PROMETHEE II method proposes the complete ranking based on the net outranking flow $\phi(a)$, so that, for all $a, b \in \mathcal{A}$,

- a is comprehensively preferred to b if $\phi(a) > \phi(b)$;
- a is indifferent with b if $\phi(a) = \phi(b)$.

In Appendix A, we describe the procedure called 'Deck of cards' to determine the weights of criteria expressing their relative importance of criteria, like in ELECTRE and PROMETHEE methods. Such weights are independent of the unit measures of criteria (scales) and must not be confused with the trade-offs.

4. Modeling interaction of criteria using non-additive measures through Choquet and Sugeno integrals

As mentioned in Section 3.1.2, the frequently used classical additive value function is based on the assumption of mutual preferential independence between any subset of criteria and its complement in $\mathcal{G}$. There are situations, however, in which this property is not satisfied. In general, it is possible that

- an evaluation on two criteria can receive a value greater than the sum of the two single criterion evaluations because of their synergy (for example, think about a car having a high maximum speed but also a good price, which is not so common and therefore there can be a bonus at the level of comprehensive evaluation),

- an evaluation on two criteria can receive a value smaller than the sum of the two single criterion evaluations because of their redundancy (for example, think about a car having a high maximum speed but also a good acceleration, which is quite common and therefore there can be a reduction at the level of comprehensive evaluation).

Aggregation procedures adopted to model synergies and redundancy between criteria are based on non-additive measures, called capacities, that assign a weight $\mu(\mathcal{H})$ to each subset of criteria $\mathcal{H} \subseteq \mathcal{G}$ which, in general, is not equal to the sum of weights of single criteria contained in set $\mathcal{H}$. The best-known such aggregation procedures are the Choquet integral for criteria expressed on a cardinal scale (Choquet, 1953), and the Sugeno integral for criteria expressed on an ordinal scale (Sugeno, 1974). Their application in the context of MCDA has been presented for the first time by Grabisch (1996). Since then, these aggregation procedures gained a great popularity for their ability of modeling complex preferences. It should be emphasized that the application of Choquet integral and Sugeno integral requires a procedure to score the performance of the criteria on a common scale and a procedure to define the capacity. The adoption of this type of procedure makes it possible to handle heterogeneous criteria, even in the case of a mixture of cardinal and ordinal criteria (see, e.g., Bottero et al., 2018). We present in detail the Choquet integral and the Sugeno integral in Appendix A.

5. Elicitation of preference information

Elicitation of preferences can be organized in a direct or indirect way. In the course of direct elicitation, the DM is asked to provide values of parameters related to a supposed form of the preference model. Experience proves that direct elicitation of numerical values of model parameters by DMs requires a great cognitive effort on their part. For example, defining weights $k_i,\ i = 1, \dots, n$, of a weighted sum model requires determination of substitution rates k_i/k_j between all pairs of criteria $\{g_i, g_j\} \subset \mathcal{G}$. These substitution rates must be, moreover, constant across the entire range of criterion scales. In the case of heterogeneous criteria, it is not easy to say what loss of performance on one criterion compensates a unit gain of performance on another criterion. Such difficulty also appears in determination of relative importance weights of criteria, having a different semantics than the weights of the weighted sum. Other preference model parameters, like indifference, preference and veto thresholds on particular criteria are not easier to elicit directly. Fishburn (1967) listed and characterized twenty-four methods of estimating additive forms of utilities.

For the above reasons, indirect preference elicitation through holistic judgments, i.e., *decision examples*, present in the literature since (Jacquet-Lagrèze & Siskos, 1982; Zionts, 1981), definitely prevails in MCDA for two decades. Decision examples are a relatively "easy" preference information. Decisions of DMs can also be observed without their active participation. For example, the choices made by people on the Internet provide a rich information source about their preferences. Moreover, psychologists confirm that DMs are more confident making decisions than explaining them (March, 1978; Slovic et al., 1977).

Decision examples can be supplied by the DM based on a collection of actual or imagined actions, or they can be derived from examining the DM's previous decisions. Methods based on the indirect elicitation of preference information are considered more user-friendly than direct elicitation approaches. Moreover, the preference model constructed on the basis of decision examples aims to be compatible with the preference information, i.e., to reproduce the holistic judgments observed in decision examples. In this way, DMs can check whether the preference model accurately reflects their preferences and what is the impact of the supplied preference information on the recommendation produced by the method. This makes the whole decision aiding process more transparent, increasing the DM's confidence in the final recommendation. As noted by Stewart (2005), MCDA includes a comprehensive process involving a rich interplay between human judgment, data analysis, and computational processes. Indeed, transparency and explainability of the decision aiding process are necessary for an efficient interplay.

Decision examples elicited in the indirect approach may have different forms depending on the problem statement and the wishes of stakeholders:

- pairwise comparisons of some real or fictitious actions represented by vectors of performances on n criteria,
- pairwise comparisons of some real actions represented by pictograms (icons) (Korhonen, 1991),
- assignments of some actions to decision classes, as their typical examples,
- intensity of preference concerning some quadruples of actions, e.g., a is preferred to b more than c is preferred to d,
- rank-related preferences, e.g., action f should be among top 5%.

The MCDA methods based on the indirect preference information elicited in the form of decision examples are called *ordinal regression methods*. They were first proposed for the additive value function preference model in the UTA method (Jacquet-Lagrèze & Siskos, 1982) that was presented in Section 3.2. The UTA method finds a *compatible* additive value function, reproducing as far as possible the preference pairwise comparisons of reference actions provided by the DM, as a result of solving an LP problem where variables are values in the breakpoints of piecewise-linear marginal value functions. While the UTA method and its many variants gained high popularity witnessed by many applications (Siskos et al., 2005), it suffers from the problem of non-uniqueness of LP solutions. In other words, the LP solution identifies a single compatible instance of the value function, while there may exist many compatible instances of the value function that reproduce the pairwise comparisons of reference actions equally well. In their original paper, Jacquet-Lagrèze and Siskos (1982) observed already that the value functions compatible with the preferences on reference actions elicited from the DM belong to a convex polyhedron. They presented various rules for managing this set. Notably, they proposed computing the minimum and maximum values attained by each marginal value function and considering their average as a final solution. However, this average is less representative in cases of large variations. In any case, different compatible instances may order non-reference actions differently, preventing a univocal recommendation. Thus, the challenge remains to account for the variability of possible preferences provided by different compatible value functions. The problem of non-univocal recommendation has been handled systematically and resolved by S. Greco, V. Mousseau and R. Słowiński in the approach called *Robust Ordinal Regression* (ROR) (Greco et al., 2008; Greco, Słowiński, Figueira, & Mousseau, 2010).

In ROR, the preference data are the same as in UTA, however, the ordinal regression identifies and uses the whole set of compatible instances of the value function. The reconstruction of the DM's judgments by the ordinal regression in terms of compatible instances of the value function is a preference learning step. Since ROR works in a loop with progressive elicitation of preferences, it is called *constructive learning* (Corrente, Greco, Kadziński, & Słowiński, 2013). In constructive learning, the model aims to reconstruct as accurately as possible the preference information of the DM, while the DM learns from the results of applying all compatible instances of the model on the considered set of actions. Constructive learning is 'robust' because it tells the truth about preferences based on a partial, not exhaustive, preference information. For example, it does not require information about all possible pairwise comparisons, like the AHP method presented in Section 3.3.

6. Rule preference model

As previously discussed, a preference model composed of "*if* ..., *then* ..." decision rules derived from decision examples, presents a dual benefit within MCDA: this approach effectively marries the ease of eliciting indirect preference information with the clarity and interpretability that decision rules afford.

The source of decision examples is twofold: they may originate from the observation of DM's historical decisions within the same context, or they might be actively solicited from the DM at the analyst's request. In the latter scenario, the examples typically pertain to specific reference actions, akin to those employed in ROR approaches. However, in both instances, it is possible for decision examples to exhibit *inconsistencies with the dominance principle* – the principle that is commonly accepted in MCDA. An inconsistency with the dominance principle may occur when:

- In case of ordinal classification: action a has been classified to a worse decision class than action b, although a is at least as good as b on all the considered criteria, i.e., a dominates b.
- In case of choice and ranking: a pair of actions (a, b) has been assigned a preference intensity smaller than pair (c, d), although differences in evaluations between a and b on all the considered criteria are at least as large as the respective differences of evaluations between c and d, i.e., pair (a, b) dominates pair (c, d).

The inconsistency can arise from various origins. For instance:

- Missing (hidden) attributes or evaluation criteria in the problem statement (Shavarani et al., 2023).
- Fluctuating preferences or hesitating DMs.

Dealing with these inconsistencies is a challenge for preference learning. They should not be considered as noise or error to be deleted from data, or mixed with consistent data by some averaging operators. They should be identified and presented through uncertain patterns.

The concept of *rough set* introduced by Pawlak (1982) appeared to be useful to handle inconsistency in data. Originally it was limited, however, to inconsistency with respect to indiscernibility principle. Actions (objects) are described by a finite set of condition attributes with nominal or cardinal scales, and a decision attribute assigning each action to one of p decision classes. Actions which have the same description by condition attributes are called *indiscernible*. The indiscernibility relation generated in the universe of discourse $\mathcal{U}$, which is the whole considered set of actions (objects) (Corcoran, 1995), constitutes the mathematical basis of rough set theory. The indiscernibility relation induces a partition of the universe into blocks of indiscernible actions, called *granules*, that can be used to build knowledge representation of a real or abstract world. Any subset X of the universe (decision class) may be expressed in terms of these blocks either precisely (as a union of granules) or approximately. In the latter case, the subset X may be characterized by two ordinary sets, called *lower approximation* and *upper approximation*. A rough set is defined by means of these two approximations, which coincide in the case of an ordinary set. The lower approximation of X is composed of all the granules included in X (whose elements, therefore, certainly belong to X), while the upper approximation of X consists of all the granules which have a non-empty intersection with X (whose elements, therefore, may belong to X). The difference between the upper and lower approximations constitutes the boundary region of the rough set, whose elements cannot be characterized with certainty as belonging or not to X by using the available information. The information about actions from the boundary region is, therefore, inconsistent or ambiguous. The cardinality of the boundary region states, moreover, the extent to which it is possible to express X in exact terms, on the basis of the available information. For this reason, this cardinality is used as a measure of vagueness of the information about X. The lower and upper approximations of a partition of $\mathcal{U}$ into decision classes prepare the ground for inducing, respectively, *certain* and *possible* knowledge patterns in the form of "*if* ..., *then* ..." decision rules. Decision rules relate some values of selected condition attributes with a value of the decision attribute.

To deal with inconsistencies violating the dominance principle, which is commonly found in preference data, S. Greco, B. Matarazzo and R. Słowiński (Greco et al., 2001) extended the original rough set concept by substituting the indiscernibility relation with a dominance relation in the definition of rough approximation of decision classes. The new methodology was called *Dominance-based Rough Set Approach* (DRSA). An important feature of DRSA is the possibility of generating the DM's preference model in terms of decision rules derived either from lower approximations of decision classes (certain rules), or from upper approximations of decision classes (possible rules), or from the difference between upper and lower approximations (approximate rules supported by inconsistent examples).

In DRSA, the granules of knowledge used to build approximations of decision classes are dominance cones in the criterion space. DRSA handles all scales of criteria because it utilizes the order only which is a primitive feature of ordinal, interval and ratio evaluation scales.

6.1. DRSA for multiple criteria classification

Let us present **DRSA to multiple criteria classification** problems first.

The finite universe of discourse $\mathcal{U}$ consists of classification examples based on reference actions from set $A^R \in \mathcal{A}$. In this case, the condition attributes are evaluation criteria, and the decision attribute assigns actions to preference-ordered decision classes. Without loss of generality, $g_i : \mathcal{U} \to \mathbf{R}$ for each $i = 1, \ldots, n$, and, for all actions $a^*, b^* \in \mathcal{U}$, $g_i(a^*) \geq g_i(b^*)$ means that "a^* *is at least as good as* b^* *on criterion* g_i", which is denoted as $a^* \succsim_i b^*$. Therefore, $\succsim_i$ is a complete preorder, i.e., a strongly complete and transitive binary relation, defined on $\mathcal{U}$ with respect to evaluations $g_i(\cdot)$. Moreover, the decision attribute d indicates to which one of p preference-ordered classes the DM has assigned a given action, such as "Poor", "Medium", or "Good". Decision attribute d makes a partition of $\mathcal{U}$ into a finite number of decision classes, called *classification*, $\mathbf{Cl}=\{Cl_1, \ldots, Cl_p\}$, such that each $a^* \in \mathcal{U}$ belongs to one and only one class Cl_t, $t = 1, \ldots, p$. Supposing that the classes are preferentially ordered, we admit that for all $r, s = 1, \ldots, p$, such that $r > s$, the actions from Cl_r are preferred to the actions from Cl_s. More formally, if $\succsim$ is a *comprehensive weak preference relation* on $\mathcal{U}$, i.e., if for all $a^*, b^* \in \mathcal{U}$, $a^* \succsim b^*$ reads "a^* *is at least as good as* b^*", then we suppose

$$[a^* \in Cl_r,\ b^* \in Cl_s,\ r > s] \Rightarrow a^* \succ b^*,$$

where $a^* \succ b^*$ means $a^* \succsim b^*$ and *not* $b^* \succsim a^*$. The assumptions presented above are typical for multiple criteria classification (also called ordinal classification with monotonicity constraints).

In DRSA, the interpretation of the assignment of actions to preference-ordered classes is based on their performances on a subset of criteria $P \subseteq \mathcal{G}$. This interpretation is called *approximation* of decision classes with respect to P. To handle the order of decision classes, they are not considered one by one but, instead, unions of classes are approximated: *upward union* from class Cl_t to class Cl_p, denoted by $Cl_t^{\geq}$, and *downward union* from class Cl_t to class Cl_1, denoted by $Cl_t^{\leq}$, i.e.:

$$Cl_t^{\geq} = \bigcup_{s \geq t} Cl_s, \quad Cl_t^{\leq} = \bigcup_{s \leq t} Cl_s, \quad t = 1, \ldots, p.$$

The statement $a^* \in Cl_t^{\geq}$ reads "a^* *belongs to at least class* Cl_t", while $a^* \in Cl_t^{\leq}$ reads "a^* *belongs to at most class* Cl_t". Let us remark that $Cl_1^{\geq} = Cl_p^{\leq} = U$, $Cl_p^{\geq}=Cl_p$ and $Cl_1^{\leq}=Cl_1$. Moreover, for t=2, ..., p, we have:

$$Cl_{t-1}^{\leq} = U \setminus Cl_t^{\geq} \quad \text{and} \quad Cl_t^{\geq} = U \setminus Cl_{t-1}^{\leq}.$$

The basic idea of the rough set concept is representation (approximation) of knowledge expressed by the decision attribute, using *granules of knowledge* generated by condition attributes. Since in

DRSA condition attributes are criteria and decision attribute indicates preference-ordered classes, the knowledge to be represented is the assignment of actions to *upward* and *downward unions of classes*, and the granules of knowledge are sets of actions belonging to *dominance cones* defined in the criterion space.

Action a^* *dominates* action b^* taking into account the criteria from set $P \subseteq \mathcal{G}$ (shortly, a^* *P-dominates* b^*), denoted by $a^* D_P b^*$, if for every criterion $g_i \in P$, $g_i(a^*) \geq g_i(b^*)$. The P-dominance relation is reflexive and transitive, i.e., it is a preorder. In general, P-dominance is not complete because, usually, there exist actions $a^*, b^* \in \mathcal{U}$ and criteria $g_i, g_{i'} \in P$, such that $g_i(a^*) > g_i(b^*)$ and $g_{i'}(b^*) > g_{i'}(a^*)$, so that neither $a^* D_P b^*$ nor $b^* D_P a^*$.

Taking into account set of criteria $P \subseteq \mathcal{G}$ and action $a^* \in U$, the granules of knowledge used in DRSA to build rough approximations are:

- set of actions dominating a^*, called *P-dominating set*, $D_P^+(a^*) = \{b^* \in U\colon b^* D_P a^*\}$,
- set of actions dominated by a^*, called *P-dominated set*, $D_P^-(a^*) = \{b^* \in U\colon a^* D_P b^*\}$.

Using multiple criteria evaluations, the above definitions can be rewritten as:

$$D_P^+\left(a^*\right) = \{b^* \in U : g_i(b^*) \geq g_i(a^*), \text{ for all } g_i \in P\},$$

$$D_P^-\left(a^*\right) = \{b^* \in U : g_i(b^*) \leq g_i(a^*), \text{ for all } g_i \in P\}.$$

Given $P \subseteq \mathcal{G}$, the inclusion of action $a^* \in U$ to the upward union of classes $Cl_t^{\geq}$, $t = 2, \ldots, p$, is *inconsistent with the dominance principle* if one of the following conditions is satisfied:

- a^* belongs to class Cl_t or better but it is P-dominated by an action b^* belonging to a class worse than Cl_t, i.e., $a^* \in Cl_t^{\geq}$ but $D_P^+(a^*) \cap Cl_{t-1}^{\leq} \neq \emptyset$,
- a^* belongs to a worse class than Cl_t but it P-dominates an action b^* belonging to class Cl_t or better, i.e., $a^* \notin Cl_t^{\geq}$ but $D_P^-(a^*) \cap Cl_t^{\geq} \neq \emptyset$.

The considerations discussed above move us closer towards a clearer understanding of rough approximations of the upward and downward unions of decision classes.

The *P-lower approximation* of $Cl_t^{\geq}$, denoted as $\underline{P}(Cl_t^{\geq})$, and the *P-upper approximation* of $Cl_t^{\geq}$, denoted as $\overline{P}(Cl_t^{\geq})$, are defined in the following way ($t = 2, \ldots, p$):

$$\underline{P}(Cl_t^{\geq}) = \{a^* \in U : D_P^+(a^*) \subseteq Cl_t^{\geq}\},$$
$$\overline{P}(Cl_t^{\geq}) = \{a^* \in U : D_P^-(a^*) \cap Cl_t^{\geq} \neq \emptyset\}.$$

The definition of *P-lower* and *P-upper approximations* of $Cl_t^{\leq}$ is analogous ($t = 1, \ldots, p-1$):

$$\underline{P}(Cl_t^{\leq}) = \{a^* \in U : D_P^-(a^*) \subseteq Cl_t^{\leq}\},$$
$$\overline{P}(Cl_t^{\leq}) = \{a^* \in U : D_P^+(a^*) \cap Cl_t^{\leq} \neq \emptyset\}.$$

The *P-boundaries* of $Cl_t^{\geq}$ and $Cl_t^{\leq}$, denoted as $Bn_P(Cl_t^{\geq})$ ($t = 2, \ldots, p$) and $Bn_P(Cl_t^{\leq})$ ($t = 1, \ldots, p-1$), respectively, are defined as: $Bn_P(Cl_t^{\geq}) = \overline{P}(Cl_t^{\geq}) \setminus \underline{P}(Cl_t^{\geq})$, $Bn_P(Cl_t^{\leq}) = \overline{P}(Cl_t^{\leq}) \setminus \underline{P}(Cl_t^{\leq})$.

For any $P \subseteq \mathcal{G}$, the actions being consistent in the meaning of the dominance principle with all upward and downward unions of classes are *P-correctly classified*. For every $P \subseteq \mathcal{G}$, the *quality of classification* **Cl** by set of criteria P is defined as the ratio of the number of P-consistent actions to the number of all actions in $\mathcal{U}$. As P-consistent actions are those which do not belong to any P-boundary $Bn_P(Cl_t^{\geq})$ or $Bn_P(Cl_t^{\leq})$, and $Bn_P(Cl_t^{\geq}) = Bn_P(Cl_{t-1}^{\leq})$ for $t = 2, \ldots, p$, the quality of classification **Cl** by criteria from set $P \in \mathcal{G}$ can be written as:

$$\gamma_P(\mathbf{Cl}) = \frac{card\left(U \setminus \left(\bigcup_{t\in\{2,\ldots,p\}} Bn_P(Cl_t^{\geq})\right)\right)}{card(U)} = \frac{card\left(U \setminus \left(\bigcup_{t\in\{1,\ldots,p-1\}} Bn_P(Cl_t^{\leq})\right)\right)}{card(U)}.$$

$\gamma_P(\mathbf{Cl})$ is interpreted as an overall consistency degree of the classification examples, with P being the considered set of criteria, and **Cl** the classification.

Any minimal subset $P \subseteq \mathcal{G}$ (in the sense of inclusion), such that $\gamma_P(\mathbf{Cl}) = \gamma_{\mathcal{G}}(\mathbf{Cl})$, is called a *reduct* of classification **Cl**, and denoted as $RED_{\mathbf{Cl}}$. Note that for a given set of classification examples one can have many reducts. The intersection of all reducts is called the *core*, and is denoted as $CORE_{\mathbf{Cl}}$. The criteria belonging to $CORE_{\mathbf{Cl}}$ are indispensable for keeping the quality of classification **Cl** at the highest level. Consequently, in set $\mathcal{G}$, there are the following categories of criteria: (i) *indispensable* criteria belonging to the core; (ii) *exchangeable* criteria belonging to some reducts, but not to the core; (iii) *redundant* criteria, neither indispensable nor exchangeable, and thus not belonging to any reduct.

The DRSA approximations of upward and downward unions of decision classes are used to induce a model of classification decisions (preference model) in the form of a set of "*if* ..., *then* ..." decision rules. Given the preference information acquired in terms of classification examples and then structured using dominance-based rough approximations, one can derive five types of decision rules referring to classification of an action $x \in \mathcal{A}$:

(1) *certain* $\mathrm{D}_{\geq}$*-decision rules*, defining lower profiles of actions from $\underline{P}(Cl_t^{\geq})$, $P = \{g_{i_1}, \ldots, g_{i_k}\} \subseteq \mathcal{G}$:
if $g_{i_1}(x) \geq r_{i_1}$ *and* ... *and* $g_{i_k}(x) \geq r_{i_k}$, *then* $x \in Cl_t^{\geq}$, $t = 2, \ldots, p$, $r_{i_1}, \ldots, r_{i_k} \in \mathbf{R}$;

(2) *possible* $\mathrm{D}_{\geq}$*-decision rules*, defining lower profiles of actions from $\overline{P}(Cl_t^{\geq})$, $P = \{g_{i_1}, \ldots, g_{i_h}\} \subseteq \mathcal{G}$:
if $g_{i_1}(x) \geq r_{i_1}$ *and* ... *and* $g_{i_h}(x) \geq r_{i_h}$, *then x possibly belongs to* $Cl_t^{\geq}$, $t = 2, \ldots, p$, $r_{i_1}, \ldots, r_{i_h} \in \mathbf{R}$;

(3) *certain* $\mathrm{D}_{\leq}$*-decision rules*, defining upper profiles of actions from $\underline{P}(Cl_t^{\leq})$, $P = \{g_{i_1}, \ldots, g_{i_k}\} \subseteq \mathcal{G}$:
if $g_{i_1}(x) \leq r_{i_1}$ *and* ... *and* $g_{i_k}(x) \leq r_{i_k}$, *then* $x \in Cl_t^{\leq}$, $t = 1, \ldots, p-1$, $r_{i_1}, \ldots, r_{i_k} \in \mathbf{R}$;

(4) *possible* $\mathrm{D}_{\leq}$*-decision rules*, defining upper profiles of actions from $\overline{P}(Cl_t^{\leq})$, $P = \{g_{i_1}, \ldots, g_{i_h}\} \subseteq \mathcal{G}$:
if $g_{i_1}(x) \leq r_{i_1}$ *and* ... *and* $g_{i_h}(x) \leq r_{i_h}$, *then x possibly belongs to* $Cl_t^{\leq}$, $t = 1, \ldots, p-1$, $r_{i_1}, \ldots, r_{i_h} \in \mathbf{R}$;

(5) *approximate* $\mathrm{D}_{\geq\leq}$*-decision rules*, defining simultaneously lower and upper profiles of inconsistent actions from $Cl_s \cup Cl_{s+1} \cup \ldots \cup Cl_t$, with no possibility of discerning to which class:
if $g_{i_1}(x) \geq r_{i_1}$ *and* ... *and* $g_{i_k}(x) \geq r_{i_k}$ *and* $g_{i_{k+1}}(x) \leq r_{i_{k+1}}$ *and* ... *and* $g_{i_h}(x) \leq r_{i_h}$, *then* $x \in Cl_s \cup Cl_{s+1} \cup \cdots \cup Cl_t$, $\{g_{i_1}, \ldots, g_{i_h}\} \subseteq \mathcal{G}$, $s, t \in \{1, \ldots, p\}$, $s < t$, $r_{i_1}, \ldots, r_{i_h} \in \mathbf{R}$.

Certain rules (type 1 and 3) represent preference patterns with full confidence factor, while possible rules (type 2 and 4) represent preference patterns with at most full (graded) confidence factor. Approximate rules (type 5) represent doubtful preference patterns. Depending on the number and quality of classification examples matching the induced rules, they are characterized by different values of some interestingness measures. Greco, Słowiński, and Szczęch (2016) recommended consideration of rule interestingness measures in four perspectives of Bayesian and likelihoodist confirmation. These measures are used in algorithms designed to classify new actions by decision rules. The algorithms, like the one presented by Błaszczyński et al. (2007), decide about classification in situations where new actions are matched by: (i) no rule, (ii) exactly one rule, (iii) several rules (even contradictory).

The definitions of rough approximations provided earlier are based on the rigorous adherence to the dominance principle. However, in

practical scenarios, especially with extensive datasets, incorporating a limited number of inconsistent examples into lower approximations proves advantageous. To address this, two parametric variations of DRSA have been introduced, each offering a distinct method for relaxing the original definitions of rough approximations: Variable Precision DRSA (Inuiguchi et al., 2009), and Variable Consistency DRSA (Błaszczyński et al., 2009). An insightful statistical interpretation of these parametric DRSA models through the lens of empirical risk minimization – a key concept to machine learning – has been carried out in (Kusunoki et al., 2021). Additionally, the development of the stochastic DRSA model, as detailed in (Kotłowski & Słowiński, 2008) and (Kotłowski et al., 2008), merits attention for its contribution to the field.

Algorithms designed for inducing decision rules from rough approximations of the unions of decision classes operate according to a variety of strategies. Among these, the "minimal-cover" strategy stands out for its efficiency, generating a concise set of rules covering all decision examples (Błaszczyński et al., 2011). This strategy employs greedy sequential covering heuristics to produce a set of rules that closely approximates minimal coverage. Other strategies focus on meeting specific user-defined criteria, such as minimal rule support, maximum rule length, or a cap on the number of rules. Additionally, some approaches aim to generate an exhaustive rule set, encompassing all rules that align with the given preference information, thereby forming a comprehensive knowledge base about the analyzed dataset. A recent development in this field involves the integration of multiple base classifiers into ensembles or committees, enhancing predictive performance through diversity (Kotłowski & Słowiński, 2009). Methods like bagging and boosting introduce diversity by altering the dataset through sampling or weighting specific examples, as noted by Błaszczyński et al. (2010) and Dembczyński, Kotłowski, and Słowiński (2010), respectively. These modified datasets are then used to train base classifiers, employing the same learning algorithm to achieve variation within the ensemble.

6.2. DRSA to multiple criteria choice and ranking

Let us next explain application of **DRSA to multiple criteria choice and ranking** problems. The classification of actions takes into account their absolute evaluations on the considered criteria, however, choice and ranking refer to pairwise comparisons of the performances of actions. This is because the rank of an action depends on its performance compared to performances of its competitors. This distinction necessitated modifying DRSA to handle pairwise comparisons of action performances instead of absolute performances typically seen in classification problems. Consequently, for choice and ranking problems, the rough approximation is addressing a comprehensive preference relation in the set of reference actions, and the approximation is made using preference relations on particular criteria for pairs of reference actions (Greco et al., 2001).

The modifications of DRSA aiming to approach choice and ranking problems are of two kinds:

(1) *Pairwise comparison table* (PCT) replaces the simple classification table: PCT is a decision table where rows represent pairs of reference actions for which performances on multiple criteria and a comprehensive preference relation are known. For example, suppose that one wants to comprehensively compare students A, B and C evaluated on two subjects (criteria), Sciences and Humanities, using an ordinal scale ("Poor", "Medium", "Good"), as shown in Table 1. The comprehensive comparison is expressed by outranking relation S, such that for any two students H and K, relation HSK is true if student H is at least as good as student K (see Table 2). The complement of S, i.e., the non-outranking relation, is marked by S^c.

Table 1
Evaluation of students with respect to Sciences and Humanities.

Student	Sciences	Humanities
A	Good	Poor
B	Poor	Good
C	Medium	Medium

Table 2
Pairwise comparison table.

Students	Sciences	Humanities	Overall comparison
(A, A)	(Good,Good)	(Poor,Poor)	S
(A, B)	(Good,Poor)	(Poor,Good)	S^c
(A, C)	(Good,Medium)	(Poor,Medium)	S
(B, A)	(Poor,Good)	(Good,Poor)	S^c
(B, B)	(Poor,Poor)	(Good,Good)	S
(B, C)	(Poor,Medium)	(Good,Medium)	S
(C, A)	(Medium,Good)	(Medium,Poor)	S^c
(C, B)	(Medium,Poor)	(Medium,Good)	S
(C, C)	(Medium,Medium)	(Medium,Medium)	S

(2) *Dominance principle* is considered for comparisons of pairs of actions instead of simple actions: if action a is preferred to b at least as strongly as c is preferred to d on all the considered criteria, then a must be comprehensively preferred to b at least as strongly as c is comprehensively preferred to d. Suppose, for example, that the DM strongly prefers evaluation "Medium" to evaluation "Poor", while weakly prefers evaluation "Good" to evaluation "Medium". Considering pairs of students (A, C) and (C, B), we have that

- A is weakly preferred to C with respect to Sciences, while C is strongly preferred to B on the same subject,
- C is strongly preferred to A with respect to Humanities, while B is weakly preferred to C on the same subject.

Thus, with respect to Sciences, the preference of C over B is stronger than the preference of A over C, while with respect to Humanities the preference of B over C is weaker than the preference of C over A, thus, if A is comprehensively at least as good as C, then it is rational to consider that also C is comprehensively at least as good as B.

The modified DRSA to the choice or ranking problems works as follows. First, the DM provides examples of pairwise comparisons of reference actions from set $A^R \in \mathcal{A}$. This information can have the form of a partial or complete ranking of these actions from best to worst. From this preference information, a PCT is constructed including an ordered pair of reference actions in each row, pairs of performances of these actions in the columns corresponding to condition attributes, and comprehensive preference relation in the column corresponding to the decision attribute. The pairs of performances for a pair of reference actions $(a^*, b^*) \in A^R \times A^R$ can be converted to graded intensities of preference in case of criteria with interval or ratio scale. Note that such a PCT with comprehensive preference relations in the decision attribute is analogous to the classification table where instead of single actions there are pairs of actions and instead of decision classes there are binary preference relations, e.g., action a^* is preferred to action b^* with a gradual intensity $\in \{+2, +1, 0, -1, -2\}$. In the simplest case, the binary preference relations can be outranking S and non-outranking S^c. Due to this analogy, all the DRSA methodology developed for multiple criteria classification can be applied to PCT and produce lower and upper approximations of the comprehensive preference relations composed of pairs of reference actions. These approximations are then used to derive certain, possible and approximate "*if* ..., *then* ..." decision rules. Variable Consistency DRSA perfectly applies to this scheme (Szeląg et al., 2014). For example, decision rules induced from

the lower approximations of binary relation S and its complement S^c can be illustrated as follows:

- *if* $g_{i_1}(x)-g_{i_1}(y) \geq \delta_{i_1}$ *and* … *and* $g_{i_k}(x)-g_{i_k}(y) \geq \delta_{i_k}$ *and* $(g_{i_{k+1}}(x) \geq r_{i_{k+1}}$ *and* $g_{i_{k+1}}(y) \leq s_{i_{k+1}})$ *and* … *and* $(g_{i_z}(x) \geq r_{i_z}$ *and* $g_{i_z}(y) \leq s_{i_z})$, *then* xSy,
  where $\{g_{i_1}, \dots, g_{i_k}\} \subseteq \mathcal{G}$ indicate criteria with a cardinal scale, $\{g_{i_{k+1}}, \dots, g_{i_z}\} \subseteq \mathcal{G}$ indicate criteria with an ordinal scale, and $\delta_{i_1}, \dots, \delta_{i_k} \in \mathbf{R}$, as well as $r_{i_{k+1}}, \dots, r_{i_z}$, $s_{i_{k+1}}, \dots, s_{i_z} \in \mathbf{R}$;
- *if* $g_{i_1}(x)-g_{i_1}(y) \leq \gamma_{i_1}$ *and* … *and* $g_{i_h}(x)-g_{i_h}(y) \leq \gamma_{i_h}$ *and* $(g_{i_{h+1}}(x) \leq r_{i_{h+1}}$ *and* $g_{i_{h+1}}(y) \geq s_{i_{h+1}})$ *and* … *and* $(g_{i_v}(x) \leq r_{i_v}$ *and* $g_{i_v}(y) \geq s_{i_v})$, *then* $xS^c y$,
  where $\{g_{i_1}, \dots, g_{i_h}\} \subseteq \mathcal{G}$ indicate criteria with a cardinal scale, $\{g_{i_{h+1}}, \dots, g_{i_v}\} \subseteq \mathcal{G}$ indicate criteria with an ordinal scale, and $\gamma_{i_1}, \dots, \gamma_{i_h} \in \mathbf{R}$, as well as $r_{i_{h+1}}, \dots, r_{i_v}$, $s_{i_{h+1}}, \dots, s_{i_v} \in \mathbf{R}$.

Such rules form a preference model of the DM. When applied to a set of new actions from $\mathcal{A}$ in view of ranking them, any pair of actions $(a, b) \in \mathcal{A} \times \mathcal{A}$ can match them in one of these ways:

- aSb and $\neg(aS^c b)$, which is *true outranking* $aS^T b$,
- $aS^c b$ and $\neg(aSb)$, which is *false outranking* $aS^F b$,
- aSb and $aS^c b$, which is *contradictory outranking* $aS^K b$,
- $\neg(aSb)$ and $\neg(aS^c b)$, which is *unknown outranking* $aS^U b$.

This four-valued outranking relation emphasizes the presence or the absence of positive or negative arguments for outranking. For any $(a, b) \in \mathcal{A} \times \mathcal{A}$ this matching result can be represented by three-valued fuzzy relation R_{3v}:

$$R_{3v}(a, b) = \frac{[aSb] + (1 - [aS^c b])}{2},$$

where $[\,\cdot\,]$ is an indicator function taking 0–1 values. A ranking rule based on R_{3v} is used to make a ranking recommendation concerning the new objects. Several ranking rules were compared by Szeląg et al. (2014) with respect to a set of desirable properties. The conclusion was that the best ranking rule is the Net Flow Rule which constructs a ranking of actions using a scoring function $NFS(a)$ for all $a \in \mathcal{A}$:

$$\begin{aligned} NFS(a) &= \sum_{b \in \mathcal{A} \setminus \{a\}} \left(R_{3v}(a, b) - R_{3v}(b, a) \right) \\ &= \sum_{b \in \mathcal{A} \setminus \{a\}} \left([aSb] - [bSa] - [aS^c b] + [bS^c a] \right), \end{aligned}$$

where $[\,\cdot\,]$ is again a 0–1 indicator function. The ranking recommendation has the form of a complete preorder on set $\mathcal{A}$ determined by $NFS(\cdot)$, and the best choice recommendation is such action $\hat{a} \in \mathcal{A}$ that $NFS(\hat{a}) = \max_{a \in \mathcal{A}} \{NFS(a)\}$.

In (Dembczyński, Kotłowski, Słowiński, & Szeląg, 2010), the application of DRSA to choice and ranking problems was compared with the statistical learning of a utility function for individual actions, employing the boosting technique. Both the Net Flow Rule and the utility function produce a linear ranking of actions. Upon concluding a computational experiment, the authors assert that both preference learning approaches – DRSA and utility function learning – exhibit commendable qualities inherent to the decision rule preference model, demonstrating robust performance in handling large datasets learning challenges.

DRSA was applied to a large spectrum of multi-attribute decision problems (Słowiński et al., 2015). Let us mention the most successful ones:

- decision-making under uncertainty and temporal preferences (Greco et al., 2010b),
- robust decision-making under uncertainty (Kadziński et al., 2016),
- classification under hierarchical structure of evaluation criteria (Dembczyński et al., 2002),
- classification under imprecise evaluations and assignments (Dembczyński et al., 2009),
- classification under missing data (Szeląg et al., 2017),
- interactive evolutionary multiobjective optimization driven by decision rules (Corrente et al., 2024).

DRSA has emerged as a widely recognized tool for organizing preference data before the induction of the preference model having the form of decision rules. It relies on simple mathematical concepts and tools, such as sets, set operations, and binary relations, without recourse to any complex algebraic or analytical frameworks (Greco et al., 2010a). This approach not only ensures that the derived decision rules are intuitively understandable, but also underscores the central principle of dominance as both logical and objective.

7. Robust Ordinal Regression (ROR)

Consider a DM aiming to rank countries based on their innovation capacity assessed by three criteria: innovation performance, direct innovation inputs, and innovation environment. To perform this ranking, a value function is selected as the preferred model. Constructing a value function that accurately reflects the DM's preferences necessitates first acquiring specific preference information from the DM. In the ROR approach, as outlined by Greco, Słowiński, Figueira, and Mousseau (2010), this typically involves the DM confidently making some pairwise comparisons among some countries that do not dominate each other, forming a reference action set $A^R = \{a^*, b^*, \dots\}$. These comparisons, such as $a^* \succeq b^*$, $b^* \succeq e^*$, and $c^* \sim d^*$, are crucial, though do not need to be exhaustive. Additionally, the DM may specify varying degrees of preference intensity between certain pairs, indicating, for instance, that the preference of a^* over b^* is stronger than that of b^* over e^*. This preference information serves as a basis for the search of the value function as the preference model. In general, it is possible that either no instance of the value function can be perfectly *compatible* with all the elicited preference information, or many instances are compatible with the preference information and are able to reproduce it perfectly. In the first case, the model cannot replicate the comprehensive judgments provided by the DM regarding the reference actions. In this scenario, the DM faces a choice: they can either proceed with a set of value function instances that are not fully consistent, or they can update certain elements of their preference information to resolve the inconsistencies. In the second case, employing ROR across all compatible instances of the value function that align with the DM's preferences allows for the derivation of two distinct preference relations across all countries, termed *necessary* and *possible*. The necessary relation holds across all compatible value function instances, whereas the possible relation is valid for at least one such instance. Additionally, ROR facilitates the determination of the extreme ranks for each country and identifies a representative instance of the value function that best matches the DM's preferences. By examining the outcomes of ROR – both the necessary and possible preference relations, along with the extreme ranks and the representative value function – the DM can gain deeper insights into their own preferences. This, in turn, may prompt the DM to introduce new preference information or modify existing one. This iterative process continues until the DM reaches a point where the necessary preference relations, extreme ranks, and/or a representative ranking collectively offer a conclusive and persuasive recommendation. Through this dynamic interaction, the DM effectively navigates through the complexities of preference modeling, ensuring the final decision is both informed and satisfactory.

Before delving into the review of ROR applications within MCDA featuring diverse preference models, it is noteworthy to mention an intriguing adaptation of ROR for decision-making under risk and uncertainty. This adaptation was feasible through the reformulation of decision-making under risk and uncertainty within the MCDA framework, where criteria are represented by significant quantiles of potential gains or losses, as detailed by Corrente, Greco, Matarazzo, and Słowiński (2016).

ROR has been adapted, moreover, to multiple criteria group decision-making (Greco et al., 2012) (a very relevant topic, as acknowledged by Raiffa et al., 2007) using two specific approaches:

- considering the set of compatible instances of the value functions that represent the preferences of all the DMs involved in the decision;
- introducing four different preference relations which are:
  - the necessary-necessary preference relation that holds when action a gets a value not worse than b by all compatible instances of value functions of all DMs in the group;
  - the necessary-possible preference relation that holds when action a gets a value not worse than b by all compatible instances of the value function of at least one DM in the group;
  - the possible-necessary preference relation that holds when action a gets a value not worse than b by at least one compatible instance of value functions of all DMs in the group;
  - the possible-possible preference relation that holds when action a gets a value not worse than b by at least one compatible instance of the value function of at least one DM in the group.

7.1. ROR with value function preference model

Let us define the ordinal regression problem utilizing the previously discussed preference information, in a similar way to the UTA method presented in Section 3.2. To represent the DM's preferences, this subsection employs a general additive value function model:

$$U(a) = \sum_{i=1}^{n} u_i(g_i(a)) \tag{6}$$

where each criterion $g_i \in \mathcal{G}$ is associated with the monotone, non-decreasing and normalized marginal value function u_i, so that the additive value (6) takes values from the interval $[0, 1]$. For notational convenience, we use $u_i(a)$ to denote $u_i(g_i(a))$, where $g_i \in \mathcal{G}$. The fundamental constraints that define the general additive instances of the value function compatible with the preference information composed of pairwise comparisons and comparisons of intensities of preference, have the following form:

$$\left.\begin{array}{l}
U(a^*) \geq U(b^*) + \varepsilon, \ \text{ if } \ a^* \succ b^*, \ \text{for } \{a^*, b^*\} \in A^R \\
U(a^*) = U(b^*), \ \text{ if } \ a^* \sim b^*, \ \text{for } \{a^*, b^*\} \in A^R \\
U(a^*) - U(b^*) \geq U(c^*) - U(d^*) + \varepsilon, \ \text{ if } \ (a^*, b^*) \\
\qquad \succ^* (c^*, d^*), \ \text{for } \{a^*, b^* c^*, d^*\} \in A^R \\
U(a^*) - U(b^*) = U(c^*) - U(d^*), \ \text{ if } \ (a^*, b^*) \sim^* (c^*, d^*), \\
\qquad\qquad\qquad\qquad \text{for } \{a^*, b^* c^*, d^*\} \in A^R \\
u_i(x_i^h) - u_i(x_i^{(h-1)}) \geq 0, \ h = 2, \ldots, m_i(A^R), \\
u_i(x_i^1) = 0, \quad \sum_{i=1}^{n} u_i(x_i^{m_i(A^R)}) = 1,
\end{array}\right\} E_{BASE}^{A^R}$$

where $(a^*, b^*) \succ^* (c^*, d^*)$ reads a^* is preferred to b^* more than c^* is preferred to d^*, $(a^*, b^*) \sim^* (c^*, d^*)$ means a^* is preferred to b^* as much as c^* is preferred to d^*, $m_i(A^R)$ is the number of different performance levels of reference actions on criterion $g_i \in \mathcal{G}$; $x_i^1, \ldots, x_i^{m_i(A^R)}$ are the ordered performances of reference actions on criterion $g_i \in \mathcal{G}$, $x_i^h < x_i^{h+1}$, $h = 1, \ldots, m_i(A^R) - 1$. The first four constraints interpret pairwise comparisons of reference actions, as well as comparisons between pairs of reference actions, and the last two constraints are monotonicity and normalization constraints.

Marginal value functions defined in this manner exhibit general monotonicity without imposing arbitrary or restrictive parameterizations (Greco, Słowiński, Figueira, & Mousseau, 2010). Note that the majority of existing ordinal regression methods employ linear or piecewise-linear marginal value functions (Siskos et al., 2005) which require specification of the number of characteristic breakpoints that is not easy for most DMs.

Observe, moreover, that due to form (6) of the additive value function, all constraints in $E_{BASE}^{A^R}$ are linear inequalities involving variables $u_i(x_i^h)$, $h = 1, \ldots, m_i(A^R)$, $g_i \in \mathcal{G}$, and ε. In order to verify if set $\mathcal{U}_{A^R}$ of compatible instances of the value function is not empty, the following LP problem with unknown variables $u_i(x_i^h)$, $h = 1, \ldots, m_i(A^R)$, $g_i \in \mathcal{G}$, and ε, has to be solved:

$$Maximize: \varepsilon, \text{ s.t. } E_{BASE}^{A^R}. \tag{7}$$

Let ε^* represent the maximum value of ε derived from solving the LP problem (7), which acts as an indicator of the margin for representation error. The set $\mathcal{U}_{A^R}$ is considered non-empty if $E_{BASE}^{A^R}$ is feasible and $\varepsilon^* > 0$. In this situation, it is possible to accurately model all given preference information using at least one instance of the value function (6). Conversely, if $E_{BASE}^{A^R}$ is infeasible or ε^* does not exceed zero, it indicates a conflict in the DM's preference information regarding the reference action set A^R, meaning that certain pairwise comparisons or intensity comparisons cannot be captured by any instance of the additive value function. In this situation, the DM is presented with three options:

- Revise conflicting preferences: Amend certain pairwise comparisons or intensity comparisons that prevent accurate modeling by an additive value function. The specific pieces of preference information causing issues can be found through solving an integer LP problem, as demonstrated by Mousseau et al. (2003).
- Accept imperfect models: Opt to proceed with a collection of value function instances within $\mathcal{U}_{A^R}$ that, although not perfectly compatible, can approximate the preference information with a margin of error not exceeding ε^*.
- Explore more complex models: Consider replacing the additive value function with a more sophisticated model, such as the Choquet integral (Angilella et al., 2010), or an additive value function augmented by components modeling positive and negative interactions between some criteria, especially when criteria vary in scale (Greco et al., 2014) – these approaches will be explored further in Section 7.5.

The exploration of numerous instances of the preference model does not guarantee a singular, clear-cut recommendation. From the viewpoint of constructive learning, where the objective shifts from prediction to the progressive revealing of preferences, it becomes crucial for users to examine the implications of utilizing the full spectrum of compatible instances from $\mathcal{U}_{A^R}$ on the action set $\mathcal{A}$. It is evident that the ultimate recommendation of a ranking, best choice, or classification can significantly differ depending on the selected instance of the model. Note, moreover, that ROR does not apply a probabilistic distribution across all compatible instances of the value function, instead treating each instance with equal validity.

In the context of evaluating a pair of actions (a, b) from set $\mathcal{A} \times \mathcal{A}$ based on the insights provided by any compatible (or non-perfectly compatible) instance of the value function from set $\mathcal{U}_{A^R}$, it is logical to assess if action a is weakly preferred over b, or the reverse, across all or at least one of the compatible instances of the value function. To address these considerations, ROR methods (Greco et al., 2008) generate two types of preference relations within the action set $\mathcal{A}$:

- *necessary* weak preference relation $\succsim^N$ that is true for a pair of actions $(a, b) \in \mathcal{A} \times \mathcal{A}$ when $U(a) \geq U(b)$ for all compatible instances of the value function,
- *possible* weak preference relation $\succsim^P$ that is true for a pair of actions $(a, b) \in \mathcal{A} \times \mathcal{A}$ when $U(a) \geq U(b)$ for at least one compatible instance of the value function.

Defined in this way, the necessary relations delineate the most reliable recommendations derived from all compatible instances of the

value function, whereas the possible relations reveal recommendations that are supported by at least one compatible instance of the value function. As such, necessary outcomes are deemed robust in light of the given preference information, ensuring that a specific relation remains valid across any compatible instance of the preference model selected for use. To ascertain the validity of both necessary and possible weak preference relations, it is essential to resolve the following programs (Corrente, Greco, Kadziński, & Słowiński, 2013):

$Maximize: \varepsilon$

s.t.

$$\left.\begin{array}{l} U(b)-U(a)\geq \varepsilon \\ E^{A^R}_{BASE} \end{array}\right\} E^N(a,b),$$

and

$Maximize: \varepsilon$

s.t.

$$\left.\begin{array}{l} U(a)-U(b)\geq 0 \\ E^{A^R}_{BASE} \end{array}\right\} E^P(a,b).$$

One concludes that $a \succsim^N b$, if $E^N(a,b)$ is not feasible or $\varepsilon_* = max\ \varepsilon$, s.t. $E^N(a,b)$, is not greater than 0, and that $a \succsim^P b$, if $E^P(a,b)$ if feasible and $\varepsilon^* = max\ \varepsilon$, s.t. $E^P(a,b)$, is greater than 0.

Note that preference relations $\succsim^N$ and $\succsim^P$ are meaningful only if set $\mathcal{U}_{A^R}$ contains at least one instance of the value function. In this case, for any $a^*, b^* \in A^R$, $a^* \succsim b^* \Rightarrow a^* \succsim^N b^*$ and $a^* \succ b^* \Rightarrow \neg(b^* \succsim^P a^*)$. In fact, if for $a^*, b^* \in A^R$ the DM says that $a^* \succsim b^*$, then for any compatible instance of the value function $U(a^*) \geq U(b^*)$, and, thus, $a^* \succsim^N b^*$. Moreover, if $a^* \succ b^*$, then for any compatible instance of the value function $U(a^*) > U(b^*)$, and, therefore, there is no compatible instance of the value function such that $U(b^*) \geq U(a^*)$, which means $\neg(b^* \succsim^P a^*)$.

The necessary weak preference relation $\succsim^N$ is a (possibly partial) preorder, i.e., it is reflexive ($a \succsim^N a$, since for all $a \in \mathcal{A}$, $U(a) = U(a)$) and transitive (for all $a, b, c \in \mathcal{A}$, if $a \succsim^N b$ and $b \succsim^N c$, then $a \succsim^N c$). Possible weak preference relation $\succsim^P$ is a strongly complete (for all $a, b \in \mathcal{A}$, $a \succsim^P b$ or $b \succsim^P a$) and negatively transitive binary relation (for all $a, b, c \in \mathcal{A}$, if $\neg(a \succsim^P b)$ and $\neg(b \succsim^P c)$, then $\neg(a \succsim^P c)$). Moreover, for any pair $(a,b) \in \mathcal{A} \times \mathcal{A}$, $a \succsim^N b \Rightarrow a \succsim^P b$, i.e., $\succsim^N \subseteq \succsim^P$, and for all $(a,b) \in \mathcal{A} \times \mathcal{A}$, $a \succsim^N b$ or $b \succsim^P a$. Axiomatic characterization of necessary and possible preferences have been provided by Giarlotta and Greco (2013).

7.2. ROR for multiple criteria classification problems

The application of ROR utilizing a general additive value function as the preference model has been extended to address multiple criteria classification problems (Greco et al., 2010a; Köksalan & Ozpeynirci, 2009). In this case, for some reference actions $a^* \in A^*$, the DM defines a desired assignment $a^* \to [C_{L^{DM}(a^*)}, C_{R^{DM}(a^*)}]$, where $C_{L^{DM}(a^*)}, C_{R^{DM}(a^*)}$ are the lowest and the highest decision class of the interval to which a^* may belong, respectively.

A single compatible instance of the value function $U \in \mathcal{U}_{A^R}$, assigns action $a \in \mathcal{A}$ to interval of classes $[C_{L^U(a)}, C_{R^U(a)}]$ in the following way:

$$L^U(a) = Max\{L^{DM}(a^*) : U(a^*) \leq U(a),\ a^* \in A^R\},$$

$$R^U(a) = Min\{R^{DM}(a^*) : U(a^*) \geq U(a),\ a^* \in A^R\}.$$

For each non-reference action $a \in \mathcal{A} \setminus A^R$ the class indices satisfy the condition $L^U(a) \leq R^U(a)$.

Given set $\mathcal{U}_{A^R}$ of compatible instances of the value function, for each action $a \in \mathcal{A}$, one can define the possible assignment $C_P(a)$ as the set of indices of classes C_h for which there exists at least one compatible instance $U \in \mathcal{U}_{A^R}$ assigning a to C_h, and the necessary assignment $C_N(a)$ as the set of indices of classes C_h for which all compatible instances $U \in \mathcal{U}_{A^R}$ assign a to C_h.

As before, in ROR for multiple criteria ranking, one can define two binary relations on the set of actions $\mathcal{A}$:

- *necessary* weak preference relation $a \succsim^N b$, if for all $U \in \mathcal{U}_{A^R}$, $U(a) \geq U(b)$,
- *possible* weak preference relation $a \succsim^P b$, if for at least one $U \in \mathcal{U}_{A^R}$, $U(a) \geq U(b)$,

for any $a, b \in \mathcal{A}$.

To compute $C_P(a)$ and $C_N(a)$, the following indices are used:

- minimum possible class:

$$L^{\mathcal{U}}_P(a) = Max\{L^{DM}(a^*) : a \succsim^N a^*,\ a^* \in A^R\}$$

- potentially minimum necessary class:

$$L^{\mathcal{U}}_N(a) = Max\{R^{DM}(a^*) : a \succsim^P a^*,\ a^* \in A^R\}$$

- potentially maximum necessary class:

$$R^{\mathcal{U}}_N(a) = Min\{R^{DM}(a^*) : a^* \succsim^P a,\ a^* \in A^R\}$$

- maximum possible class:

$$R^{\mathcal{U}}_P(a) = Min\{R^{DM}(a^*) : a^* \succsim^N a,\ a^* \in A^R\}$$

The possible and necessary assignments $C_P(a)$ and $C_N(a)$ are thus defined as:

$$C_P(a) = [L^{\mathcal{U}}_P(a), R^{\mathcal{U}}_P(a)]$$

and, if $L^{\mathcal{U}}_N(a) \leq R^{\mathcal{U}}_N(a)$, then

$$C_N(a) = [L^{\mathcal{U}}_N(a), R^{\mathcal{U}}_N(a)],$$

otherwise $C_N(a) = \emptyset$.

Proceeding analogously to Figueira et al. (2009), one can handle within ROR for multiple criteria classification an additional preference information about intensity of preference between pairs of reference actions.

An alternative approach to multiple criteria classification problems considering multiple compatible value functions have been presented in (Köksalan & Ulu, 2003) and (Ulu & Köksalan, 2014). This approach consists of the interactive placement of actions into preference-ordered classes by a DM, assuming that the driving preference model is a linear or quasiconcave value function.

7.3. Extreme ranking analysis in ROR with value function

As different compatible instances of the value function may assign a different position to a given action in the ranking of all actions from set $\mathcal{A}$, it is interesting to identify its highest and lowest ranks, as well as the score that an action can attain. This procedure is known in ROR as Extreme Ranking Analysis (Kadziński et al., 2012a; Köksalan et al., 2010). To identify the range of ranks $[P^*(a), P_*(a)]$ a particular action $a \in \mathcal{A}$ could attain due to multiple compatible (or almost compatible) instances of the value function, the following mixed-integer LP problems have to be solved:

$$Minimize:\ f^{rank}_{max} = \sum_{b \in \mathcal{A}\setminus\{a\}} v_b$$

s.t.

$$\left.\begin{array}{ll} U(a)-U(b)+Mv_b \geq \varepsilon, & \text{for all } b \in \mathcal{A}\setminus\{a\} \\ E^{A^R}_{BASE} & \end{array}\right\} E^{A^R}_{max},$$

and

$$Minimize:\ f^{rank}_{min} = \sum_{b \in \mathcal{A}\setminus\{a\}} v_b$$

s.t.

$$\left.\begin{array}{ll} U(b)-U(a)+Mv_b \geq \varepsilon, & \text{for all } b \in \mathcal{A}\setminus\{a\} \\ E^{A^R}_{BASE} & \end{array}\right\} E^{A^R}_{min},$$

where M is a big positive constant value, ε is an infinitesimal positive constant value, and v_b is a binary variable associated with comparison of action a to b. Note that in these two problems there are $card(\mathcal{A})-1$ binary variables, each corresponding to $b \in \mathcal{A}\backslash\{a\}$. One concludes that:

$$P^*(a) = f^{rank}_{max} + 1 \quad \text{and} \quad P_*(a) = card(\mathcal{A}) - f^{rank}_{min}.$$

One may also identify the ranges of comprehensive values $[U_*(a), U^*(a)]$ that a particular action a could attain. To get these values, one has to minimize and maximize $U(a)$, subject to $E^{A^R}_{BASE}$, respectively.

7.4. Stochastic multiobjective acceptability analysis in ROR with value function

Since the necessary preference relation $\succsim^N$ is a partial preorder, in general, certain pairs of actions remain incomparable, that is, neither $a \succsim^P b$ nor $b \succsim^P a$ can be conclusively determined. To estimate the probability that a randomly chosen compatible instance of the value function would rank action a higher than action b, or the reverse, one can utilize Stochastic Multiobjective Acceptability Analysis (SMAA) (Kadziński & Tervonen, 2013). This method involves a *Hit & Run* sampling technique for selecting compatible instances of the value function within constraints $E^{A^R}_{BASE}$ and then assessing the frequency of occurrences where:

- $a \succ b$, yielding a pairwise winning index $p(a,b)$,
- a is assigned position h in the ranking, resulting in a rank acceptability index r^h_a.

In the SMAA framework, each compatible instance of the value function is considered to have an equal probability of occurrence. However, Corrente, Greco, Kadziński, and Słowiński (2016) suggested incorporating a probability distribution over the compatible instances of the value function based on additional, uncertain preference information, such as the belief that $a \succ b$ is more plausible than $b \succ a$. This approach is structured around three main steps:

- sampling a specified number (s_v) of value function instances $U_t \in \mathcal{U}_{A^R},\ t = 1,\dots,s_v$, that align with the certain preference information provided by the DM,
- generating a probability distribution $\{w_t(U_t) \in [0,1] : \sum_{t=1}^{s_v} w_t(U_t) = 1\}$ for the sampled instances, incorporating the uncertain preference information through LP to weight each instance,
- sampling a number of probability distributions that accord with the uncertain preference information and identifying the barycenter (centroid) distribution to serve as a representative distribution for establishing the final ranking.

7.5. ROR with non-additive value functions

When DMs show preferences that involve *interactions* among criteria, the general additive model may fall short in representing them. Interactions can manifest as either *negative synergy*, where the combined importance of a group of criteria is less than the sum of their individual weights, or *positive synergy*, where the combined weight exceeds the sum of individual weights.

To address the complexity of criteria interactions, *non-additive integrals* can be employed as preference models. Specifically, the Choquet integral is suitable for cardinal criteria (quantifiable on interval or ratio scales), while the Sugeno integral is suited for ordinal criteria (Grabisch, 1996). These integrals offer a refined approach to capturing the intricacies of DM preferences by accounting for interactions among criteria. For their detailed presentation, see Appendix B.

There is, however, an important requirement when applying these integrals: all criteria evaluations must be expressed on the same normalized scale. This condition is often difficult to meet in the complex scenarios encountered in MCDA (Roy, 2009).

The condition of the same scale has been overcome in the UTA^{GMS}-INT method (Greco et al., 2014). The authors introduced a novel method for aggregating interacting criteria with diverse scales, using a value function that combines the sum of marginal non-decreasing value functions $u_i(a),\ i = 1,\dots,n$, with additional sums of functions:

$$syn^+_{i_1,i_2} : [x^{min}_{i_1}, x^{max}_{i_1}] \times [x^{min}_{i_2}, x^{max}_{i_2}] \to [0,1],$$

$$syn^-_{i_1,i_2} : [x^{min}_{i_1}, x^{max}_{i_1}] \times [x^{min}_{i_2}, x^{max}_{i_2}] \to [0,1], \quad \text{for all} \quad (g_{i_1}, g_{i_2}) \in \mathcal{G}\times\mathcal{G},\ i_1 > i_2,$$

where $x^{min}_{i_1}, x^{max}_{i_1}$ and $x^{min}_{i_2}, x^{max}_{i_2}$ are, respectively, the worst (min) and the best (max) performances on criteria $g_{i_1}, g_{i_2} \in \mathcal{G}$. Functions $syn^+_{i_1,i_2}(x_{i_1}, x_{i_2})$ and $syn^-_{i_1,i_2}(x_{i_1}, x_{i_2})$, representing respectively, positive and negative interactions between two criteria $(g_{i_1}, g_{i_2}) \in \mathcal{G}\times\mathcal{G}$, such that $i_1 > i_2$, increase or decrease the value function's additive component. They act as *bonus* and *penalty* functions, respectively, and are non-decreasing in their two arguments. Given that pairs of criteria can exhibit either positive or negative synergy, $syn^+_{i_1,i_2}(\cdot,\cdot)$ and $syn^-_{i_1,i_2}(\cdot,\cdot)$ are mutually exclusive.

For action $a \in \mathcal{A}$, the value function is defined as:

$$U^{int}(a) = \sum_{i=1}^{n} u_i(a) + \sum_{(g_{i_1},g_{i_2})\in\mathcal{G}\times\mathcal{G},\ i_1>i_2} syn^+_{i_1,i_2}(g_{i_1}(a), g_{i_2}(a)) - \sum_{(g_{i_1},g_{i_2})\in\mathcal{G}\times\mathcal{G},\ i_1>i_2} syn^-_{i_1,i_2}(g_{i_1}(a), g_{i_2}(a)).$$

The U^{int} model comprehensively captures both positive and negative interactions among criteria in pairs, resembling the 2-additive Choquet integral but with enhanced preference representation capabilities, as demonstrated in (Greco et al., 2014). In practice, U^{int} might be even overly general, so it is worthwhile to identify a minimal number of criteria that interact in pairs.

7.6. ROR with outranking relation preference model

ROR can be applied also to outranking methods presented in Section 3.9, providing a very useful support in fixing the many parameters requested by these methods (indifference, preference and veto thresholds, weights of criteria, concordance threshold, and so on). In this scenario, the DM communicates their preferences through holistic pairwise comparisons between specific reference actions: $a^* S b^*$ or $a^* S^c b^*$ for some $a^*, b^* \in A^R \subseteq \mathcal{A}$, where S^c represents the negation of the outranking relation. Furthermore, as explained in (Greco, Kadziński, Mousseau, & Słowiński, 2011) where adaptation of ROR to ELECTRE methods is discussed, the DM can define ranges $[q_{i*}, q^*_i]$ and $[p_{i*}, p^*_i]$, which correspond to the limits of indifference and preference thresholds, respectively, that the DM considers acceptable.

An outranking model compatible with preference information provided by the DM is a set of preference model parameters reproducing the DM's preference information concerning pairs $(a^*, b^*) \in A^R \times A^R$.

For any pair of actions $(a,b) \in \mathcal{A}$ we say that *a necessarily outranks b*, denoted by $aS^N b$, if a outranks b for all compatible instances of the outranking model, while we say *a possibly outranks b*, denoted by $aS^P b$, if a outranks b for at least one compatible instance of the outranking model.

To formulate a recommendation for a choice problem, it is recommended to identify a kernel within the necessary outranking graph S^N. For ranking problems, the Net Flow Score (NFS) procedure applied to the outranking graph, which encompasses both S^N and S^P relations, is suitable. This method calculates for each action $a \in \mathcal{A}$, a score $NFS(a) = strength(a) - weakness(a)$. Here, $strength(a)$ tallies all relations (arcs) where action a outranks or does not get outranked by other actions z, captured as $aS^N z$ or *not* $zS^P a$. Conversely, $weakness(a)$ accounts for all relations where action a is outranked or does not outrank other actions v, represented as $vS^N a$ or *not* $aS^P v$. The outcome is a complete preorder of actions based on $NSF(a)$ for all $a \in \mathcal{A}$.

Other developments in ROR for outranking methods applied to multiple criteria ranking include:

- ROR for PROMETHEE method and extreme ranking analysis (Kadziński et al., 2012a),
- representative instance of compatible outranking relations (Kadziński et al., 2012b).

7.7. ROR handling the hierarchy of criteria

In numerous complex decision-making scenarios, the set of criteria $\mathcal{G}$ often possesses a *hierarchical* structure. This implies that the performance on certain criteria is derived from the aggregated performances of their sub-criteria. As noted in (Keeney & Raiffa, 1976): "Almost everyone who has seriously thought about the objectives in a complex problem has come up with some sort of hierarchy of objectives". This perspective is echoed in (Belton & Stewart, 2002), where the authors write: "In the process of structuring the problem, it is possible (even likely) that the criteria may have been constructed hierarchically in terms of a value tree". An instance of this is observed in the land use problem discussed in (Belton & Stewart, 2002), where the economic benefit criterion encompasses three sub-criteria: outputs from agriculture, forestry, and secondary industry.

Considering hierarchical criteria not only simplifies the problem complexity by allowing analysis at various levels of specificity but also facilitates a more systematic evaluation process. By constructing a hierarchy, DMs can initially assess actions based on elementary criteria (the leaves of the hierarchy tree) and subsequently understand their broader implications on higher-level criteria. This methodical approach aids in achieving a more nuanced understanding of the decision-making process, enhancing the quality and coherence of the recommendations.

The ROR method has been tailored to address the hierarchical structure of criteria through the Multiple Criteria Hierarchy Process (MCHP). Initially, this adaptation was implemented for ranking problems and a general additive value function as the preference model (Corrente et al., 2012).

In MCHP, the DMs can articulate their preferences for reference actions either holistically or in relation to specific macro- or sub-criteria within the hierarchy. This dual approach enables a more accurate expression and subsequent representation of their preferences. Specifically, MCHP provides both necessary and possible preference relations not only at the comprehensive level but also concerning each distinct macro- or sub-criterion in the hierarchy, enhancing the precision and relevance of preference modeling.

Since its initial introduction, the ROR-MCHP approach with a general additive value function for ranking problems has seen significant extensions. They have diversified its application across various preference modeling frameworks, including:

- The adoption of an outranking relation preference model, as utilized in the ELECTRE and PROMETHEE methods, for both multiple criteria ranking (Corrente, Greco, & Słowiński, 2013) and classification (Corrente, Doumpos, et al., 2017; Corrente, Greco, & Słowiński, 2016).
- The implementation of an outranking relation within the ELECTRE III method, tailored for multiple criteria ranking, addressing the challenge of imprecise criteria weights, and incorporating stochastic multiobjective acceptability analysis (Corrente, Figueira, et al., 2017).
- The integration of the Choquet integral as a preference model for multiple criteria ranking, again leveraging stochastic multiobjective acceptability analysis (Angilella et al., 2016).

7.8. ROR applied to the rule preference model

In (Kadziński et al., 2015), ROR was adapted to multiple criteria classification with the rule preference model, incorporating all minimal-cover (MC) sets of decision rules aligned with the provided decision examples to formulate recommendations for non-reference actions. An MC set represents a single compatible instance of the preference model, encompassing a unique selection of rules. The process begins by inducing a comprehensive set of potential rules, followed by the identification of all compatible MC sets through solving a series of integer LP problems. This ensures each reference action is covered by at least one rule from every MC set. The resulting ensemble of MC classifiers is then applied to all actions under consideration, yielding two categories of classification outcomes: action a is *necessarily* assigned to class Cl_h if it is consistently placed in Cl_h by all MC sets, and it is *possibly* assigned to Cl_h, if at least one MC set assigns it to that class.

Furthermore, this methodology allows for the calculation of class acceptability indices, reflecting the proportion of MC sets of rules that assign an action to a specific class or a group of adjacent classes. The robustness of these classifications is examined through a dual approach. Initially, a representative MC set, which mirrors the most stable aspects of the classification based on acceptability indices, is selected. Subsequently, for each action, rules that significantly impact the recommendation across the largest number of compatible MC sets are identified. These pivotal rules offer valuable insights into the classification logic, aiding DMs in understanding the rationale behind the recommendations.

8. Conclusions

In this article, we have reviewed the developments of Multiple Criteria Decision Analysis (MCDA) over a period of last fifty years. Our focus has been on the case of a finite set of potential actions. We have described many "classic" methods of handling such problems, including the value/utility function assessment, the outranking approach, and the Analytic Hierarchy Process. A large part of our paper described Robust Ordinal Regression which underlines the current trend in MCDA. While Robust Ordinal Regression integrates many of the previous developments in the field, it implements the paradigm of preference learning from holistic preference information (decision examples), typical for Artificial Intelligence (AI).

Despite the great progress which has been made in our field over the reviewed period, major challenges and opportunities still lie ahead. We believe that the future for our field is bright. Considering the space for the more or less explicit application of MCDA, we observe that in an increasingly complex world, the ability to adequately consider many different points of view is certainly the fundamental prerequisite for handling operational decision problems in any domain. Moreover, the scoring of units of different natures on the basis of a plurality of elementary indices is becoming pervasive in an increasing number of domains, as in the case of the ranking of universities (Liu & Cheng, 2005), well-being indices (Rowan, 2023), ESG assessments (Dorfleitner et al., 2015), and so on. This is the domain of so-called *composite indices* (Greco et al., 2019) which are basically multiple criteria evaluations. Therefore, the demand for multiple criteria decision analysis is constantly growing, and we believe that adopting MCDA methodologies, procedures, and techniques is necessary to successfully address the complex problems faced by our society. Users of scientific decision-support methods are also increasingly aware that decision problems involving multiple evaluation criteria require efficient methods of preference learning from big data describing decision situations (Hüllermeier & Słowiński, 2024).

The technological and other mega trends facing the world offer great opportunities for our field (Wallenius & Wallenius, 2022), but we must seize the opportunities. Related to the mega trends, we see great potential for novel contributions, for example, in the following areas:

1. Developing better recommender systems and search engines.
2. Developing better matching algorithms for various situations.
3. Promoting the use of multivariate big data in companies and the public sector.

4. Developing better measures (indices) for sustainable investing in the spirit of Hallerbach et al. (2004).
5. Developing decision support tools targeted at masses to help consumers make online purchasing more efficient.
6. Theoretical and applied work on conflict resolution.
7. Work on risk analysis, and improving the preparedness of organizations, countries, and regions for eventual crises.

Most, if not all of the above challenges or opportunities are centrally connected to preference modeling, which is a core concept in our field. Besides the technological mega trends providing research and application opportunities for our field, we foresee several other trends.

We believe that it is essential for our field to better understand the behavioral realities. Our field is not only about mathematics, but also about human psychology. Integrating method development better with decision psychology presents many future research opportunities. One interesting approach is to base conclusions about the cognitive load on human DMs imposed by various decision methods on neurophysiological measurements, such as eye movements, skin reactions, facial expressions, EEG data. For an early attempt, see Ravaja et al. (2016).

Modern computer technology offers opportunities for various visualizations and visual aids. We see progress in visually representing problems (also in higher dimensions) and eliciting preference information in visual format.

Heuristics are increasingly being used and studied, in particular, in case of multiple criteria decision problems with very many actions. In such cases, it becomes less important to achieve "optimal" rather than "good enough" (or satisficing, using the term coined by Herbert Simon) solutions. At the other end there are multiple criteria decision problems where generating a set of potential actions is a hard optimization or simulation problem. Such situations will also need smart heuristics.

Moreover, as predicted by Peter Fishburn already some 30 years ago, the role of transitivity (which has played an important role in our field) slowly erodes in Multiple Criteria Decision Analysis (Fishburn, 1991). The evolution of our field also occurs as a result of changes in the nature of users, i.e., clients of our scientific services. Fifteen years ago, (Wallenius et al., 2008) made an observation that remains valid today: *Decision support in a distributed environment is somewhat different from what we had expected in the early 1990's. For example, what is a user? Our field has traditionally sought to support corporate managers. However, household consumers need support for purchasing decisions. What kind of decision support do they want in an Internet or mobile environment? The problem may not be one of having insufficient information, but rather one of having "too much" or an unknown quality of information. We may have to filter information.* This is still a potential application and development area for MCDA.

Over the past fifty years, the application area of MCDA methods was systematically growing. This is witnessed by existence in the MCDA literature of a large number of methods adapted to various applications. All symptoms indicate that this trend will continue. Facing a new decision problem involving multiple evaluation criteria, we advise to check first if there already exists an MCDA method which could be properly used in the context of the considered application. Some guidelines for this check have been provided by Roy and Słowiński (2013), and then enhanced by Cinelli et al. (2022) based on a specially created taxonomy that takes into account the decision problem characteristics, the DM's expectations for the form of the recommendation, the type of preference information that can be obtained from the DM, and desired features of the applied preference model. This taxonomy has been implemented in the Multiple Criteria Decision Analysis Methods Selection Software (MCDA-MSS) made available online at https://mcda.cs.put.poznan.pl/. It includes a big collection of (> 200) MCDA methods.

Finally, as is already evident today, AI offers great opportunities for many fields, also for preference-driven decision support. Perhaps in the future, the DMs could provide a verbal description of what they want and consider important, instead of being asked to respond to precise preference questions (in terms of criteria). AI could, perhaps, be used to interpret what the DMs mean (in terms of the importance of criteria and objectives), and what kind of actions (or solutions) would be of interest to them. After seeing a tentative list of potentially interesting actions (or solutions), the DMs could update their verbal description concerning what they want. We believe this could be accepted by users if in addition to receiving recommendations from an "intelligent" decision support system, users would be provided with a comprehensive explanation. Furthermore, we think that engaging users in a meaningful exchange of arguments with the system would be necessary to arrive at a truly satisfying decision.

CRediT authorship contribution statement

Salvatore Greco: Investigation, Methodology, Writing – original draft, Writing – review & editing. **Roman Słowiński:** Investigation, Methodology, Writing – original draft, Writing – review & editing. **Jyrki Wallenius:** Investigation, Methodology, Writing – original draft, Writing – review & editing.

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Appendix A. The 'deck of the cards' method: a procedure to determine weights of ELECTRE and PROMETHEE methods

The weights used in the ELECTRE and PROMETHEE methods are expressing the relative importance of criteria. They are independent of the unit measures of the criteria and must not be confused with the trade-offs used in MCDA methods based on value function preference model. An operational method to determine these weights uses the analogy of a 'deck of the cards' – it was proposed by Simos (1990) and later improved by Figueira and Roy (2002). The 'deck of the cards' method determines the weights of criteria in the following steps:

- the DM is given a deck of cards composed of cards with names of particular criteria $g_i \in \mathcal{G}$ and some blank cards;
- the DM is asked to rank order the cards-criteria according to the relative importance of criteria, starting from the least important to the most important; cards-criteria with the same importance have to be put in the same equivalence class $\mathcal{G}_h$; the order of the equivalence classes $\mathcal{G}_1, \dots, \mathcal{G}_p$ is such that $\mathcal{G}_1 \cup \dots \cup \mathcal{G}_p = \mathcal{G}$, and criteria from $\mathcal{G}_h$ are more important than criteria from $\mathcal{G}_{h-1}$, $h = 2, \dots, p$;
- the DM is asked to put some blank cards between the equivalence classes $\mathcal{G}_h$ and $\mathcal{G}_{h+1}$ to amplify the difference of the importance of criteria belonging to $\mathcal{G}_h$ and to $\mathcal{G}_{h+1}$, $h = 1, \dots, p-1$; let us denote by e_h the number of blank cards placed between $\mathcal{G}_h$ and $\mathcal{G}_{h+1}$; the greater e_h, the greater the difference of importance between criteria from $\mathcal{G}_h$ and $\mathcal{G}_{h+1}$;
- the DM is asked for the value z indicating how many times the weight of the most important criteria, that is, criteria from $\mathcal{G}_p$, is greater than the weight of the least important criteria, that is, criteria from $\mathcal{G}_1$;
- on the basis of this preferential information, each criterion $g_i \in \mathcal{G}$ is assigned weight w_i, such that

– if $g_{i_1} \in \mathcal{G}_h$ and $g_{i_2} \in \mathcal{G}_{h+1}$,

$$w_{i_2} = w_{i_1} + (e_h + 1) \times c, \tag{A.1}$$

with $c > 0$ representing the value of a card,

– if $g_{i_1} \in \mathcal{G}_1$ and $g_{i_2} \in \mathcal{G}_p$, then

$$w_{i_2} = z \times w_{i_1}; \tag{A.2}$$

– and

$$\sum_{g_i \in \mathcal{G}} w_i = 1. \tag{A.3}$$

Applying iteratively (A.1), and denoting by w^h the weight of a criterion in equivalence class $\mathcal{G}_h$, $h = 1, \dots, p$, we get

$$w^2 = w^1 + (e_1 + 1) \times c, \quad w^3 = w^2 + (e_2 + 1) \times c = w^1 + (e_1 + e_2 + 2) \times c, \ \dots$$

and, in general,

$$w^h = w^1 + \left(\sum_{k=1}^{h-1} e_k + h - 1 \right) \times c. \tag{A.4}$$

Denoting by n_h the number of criteria in $\mathcal{G}_h$, $h = 1, \dots, p$, so that $\sum_{h=1}^{p} n_h = n$, (A.3) and (A.4) give

$$n \times w^1 + \sum_{h=2}^{p} n_h \times \left(\sum_{k=1}^{h-1} e_k + h - 1 \right) \times c = 1. \tag{A.5}$$

Observe that (A.2) can be rewritten as $w^p = z \times w^1$, so that, taking into account (A.4) we get

$$w^1 + \left(\sum_{k=1}^{p-1} e_k + p - 1 \right) \times c = z \times w^1. \tag{A.6}$$

From (A.5) and (A.6), we get the following values of w^1 and c:

$$w^1 = \frac{\sum_{k=1}^{p-1} e_k + p - 1}{n \times \left(\sum_{k=1}^{p-1} e_k + p - 1 \right) + \sum_{h=2}^{p} n_h \times \left(\sum_{k=1}^{h-1} e_k + h - 1 \right) \times (z - 1)} \tag{A.7}$$

$$c = \frac{z - 1}{n \times \left(\sum_{k=1}^{p-1} e_k + p - 1 \right) + \sum_{h=2}^{p} n_h \times \left(\sum_{k=1}^{h-1} e_k + h - 1 \right) \times (z - 1)}. \tag{A.8}$$

Consequently, we can say that, operationally, once the DM has placed e_h blank cards between equivalence classes $\mathcal{G}_h$, $h = 1, \dots, p$, and provided the value of z, using (A.7) and (A.8) one can determine the values of w^1 and c, and then, using (A.4), the weights of all particular criteria from $\mathcal{G}$.

To illustrate how the method works, let us consider the following example. Suppose, the set of criteria is composed of eight criteria $\mathcal{G} = \{g_1, g_2, g_3, g_4, g_5, g_6, g_7, g_8\}$. The DM orders the criteria from the least important to the most important putting a number of blank cards (shown in square brackets) between equivalence classes as follows:

$$\mathcal{G}_1 = \{g_3, g_6\} \ [4] \ \mathcal{G}_2 = \{g_1\} \ [0] \ \mathcal{G}_3 = \{g_2, g_4, g_7\} \ [3] \ \mathcal{G}_4 = \{g_5, g_8\}.$$

Thus, we have $e_1 = 4, e_2 = 0, e_3 = 3$. The DM requires, moreover, that the weight of criteria from $\mathcal{G}_4$ has a value 10 times greater than the weight of criteria from $\mathcal{G}_1$, that is, $z = 10$. Applying (A.7) and (A.8), we get $w^1 = 0.021$ and $c = 0,019$, so that, by (A.4), we get

$$w^2 = w^1 + (e_1 + 1) \times c = 0.118, \quad w^3 = w^2 + (e_2 + 1) \times c = 0.137,$$
$$w^4 = w^3 + (e_3 + 1) \times c = 0.214,$$

and, consequently, $w_3 = w_6 = w^1 = 0.021$, $w_1 = w^2 = 0.118$, $w_2 = w_4 = w_7 = w^3 = 0.137$, $w_5 = w_8 = w^4 = 0.214$.

Let us conclude this appendix by mentioning that Siskos and Tsotsolas (2015) considered a reformulation of the Simos method that provides a non-empty convex polyhedral set of weight vectors, proposing a set of complementary robustness analysis rules and measures integrated in the Robust Simos Method.

Appendix B. Choquet integral and other non-additive integrals

The Choquet integral is an aggregation function defined with respect to a capacity (also called fuzzy measure) which assigns a weight not to each single criterion, but to all subsets of criteria from $\mathcal{G}$. Formally, a capacity μ is a function assigning a value between 0 and 1 to each subset of criteria $\mathcal{H} \subseteq \mathcal{G}$, such that the following conditions are satisfied:

- boundary conditions: $\mu(\emptyset) = 0$ and $\mu(\mathcal{G}) = 1$,
- monotonicity: if $\mathcal{H} \subseteq \mathcal{K} \subseteq \mathcal{G}$, then $\mu(\mathcal{H}) \leq \mu(\mathcal{K})$.

Assuming that the evaluations on all criteria from $\mathcal{G}$ are normalized between 0 and 1, representing the worst and the best possible performance, respectively, for each $\mathcal{H} \subseteq \mathcal{G}$, $\mu(\mathcal{H})$ is interpreted as a comprehensive evaluation of an action having evaluation 1 on criteria from $\mathcal{H}$ and evaluation 0 on the remaining criteria. In general, on the basis of capacity μ, the Choquet integral gives the following comprehensive evaluation to action $a \in \mathcal{A}$:

$$Ch(a) = \sum_{i=1}^{n} (g_{(i)}(a) - g_{(i-1)}(a)) \times \mu(\{g_j \in \mathcal{G} : g_j(a) \geq g_{(i)}(a)\}), \tag{B.1}$$

with $g_{(1)}(a) \leq \dots \leq g_{(n)}(a)$ being the evaluations of action a on criteria from $\mathcal{G}$ ordered from the lowest to the highest and $g_{(0)}(a) = 0$.

It is interesting to observe that if there are no interactions between criteria – in the sense that for all $\mathcal{H}, \mathcal{K} \subseteq \mathcal{G}$, such that $\mathcal{H} \cap \mathcal{K} = \emptyset$, $\mu(\mathcal{H} \cup \mathcal{K}) = \mu(\mathcal{H}) + \mu(\mathcal{K})$ – for all subsets of criteria $\mathcal{H} \subseteq \mathcal{G}$, putting $w_i = \mu(\{g_i\}), i = 1 \dots, n$, we have

$$\mu(\mathcal{H}) = \sum_{g_i \in \mathcal{H}} w_i$$

and taking into consideration the weighted sum formulation of value function we get

$$WS(a) = \sum_{i=1}^{n} w_i g_i(a) = \sum_{i=1}^{n} (g_{(i)}(a) - g_{(i-1)}(a)) \times \sum_{g_j \in \mathcal{G}: g_j(a) \geq g_{(i)}(a)} w_j =$$

$$= \sum_{i=1}^{n} (g_{(i)}(a) - g_{(i-1)}(a)) \times \mu(\{g_j \in \mathcal{G} : g_j(a) \geq g_{(i)}(a)\}) = Ch(a).$$

Considering for each $a \in \mathcal{A}$ function $F_a : [0, 1] \to \mathbb{R}$ defined as

$$F_a(x) = g_i(a) \quad \text{for} \quad \sum_{j=1}^{i-1} w_j < x \leq \sum_{j=1}^{i} w_j,$$

we can interpret $WS(a)$ as the area of the set $S(a) = \{(x, y) : 0 \leqslant x \leqslant 1, 0 \leqslant y \leqslant F_a(x)\}$, that is the surface between x axis and the graph of function $F_a : [0, 1] \to \mathbb{R}$. Indeed, observe that $S(a)$ is the union of the rectangles $R(a, i), g_i \in \mathcal{G}$, having as base the interval $\left[\sum_{j=1}^{i-1} w_j, \sum_{j=1}^{i} w_j\right]$ on the x-axis and a height measuring $g_i(a)$. Each rectangle $R(a, i), g_i \in \mathcal{G}$, has an area $\rho(R(a, i)) = w_i \times g_i(a)$, so that for the comprehensive area of $S(a)$ we get

$$\rho(S(a)) = \sum_{i=1}^{n} \rho(R(a, i)) = \sum_{i=1}^{n} w_i \times g_i(a) = WS(a).$$

Let us observe now that we can obtain another set $S^*(a)$ with the same area of $S(a)$ by considering function $F_a^* : [0, 1] \to \mathbb{R}$ defined as

$$F_a^*(x) = g_{(i)}(a) \quad \text{for} \quad \sum_{j=1}^{i-1} w_{(j)} < x \leq \sum_{j=1}^{i} w_{(j)}$$

and set $S^*(a) = \{(x, y) : 0 \leqslant x \leqslant 1, \ 0 \leqslant y \leqslant F_a^*(x)\}$, that is the surface between x axis and the graph of function $F_a^* : [0, 1] \to \mathbb{R}$. Indeed, set $S^*(a)$ is obtained from set $S(a)$ by permutation of the rectangles $R(a, i)$ that are reordered from the left to the right in the order $(1), \dots, (n)$. Observe that we can decompose "horizontally" set $S^*(a)$ in the union of rectangles $R^*(a, (i))$ having

- as a base, the segment with extreme points $P_{(i)} \equiv \left(0, g_{(i-1)}(a)\right)$ and $Q_{(i)} \equiv \left(\sum_{j=1}^{(i)} w_{(j)}, g_{(i-1)}(a)\right)$ and length $\mu\left(\{g_j \in \mathcal{G} : g_j(a) \geq g_{(i)}(a)\}\right) = \sum_{j=1}^{i} w_{(j)}$, and
- a height measuring $g_{(i)}(a) - g_{(i-1)}(a)$.

Each rectangle $R^*(a,(i)),\ i = 1, \dots, n$, has an area

$$\rho\left(R^*(a,(i))\right) = \mu\left(\{g_j \in \mathcal{G} : g_j(a) \geq g_{(i)}(a)\}\right) \times \left(g_{(i)}(a) - g_{(i-1)}(a)\right) = \sum_{j=1}^{i} w_{(j)} \times \left(g_{(i)}(a) - g_{(i-1)}(a)\right),$$

so that for the comprehensive area of $S^*(a)$ we get

$$\rho\left(S^*(a)\right) = \sum_{i=1}^{n} \rho\left(R^*(a,(i))\right) = \sum_{i=1}^{n} \mu\left(\{g_j \in \mathcal{G} : g_j(a) \geq g_{(i)}(a)\}\right) \times \left(g_{(i)}(a) - g_{(i-1)}(a)\right) = \sum_{i=1}^{n} \left(\left(\sum_{j=1}^{i} w_{(j)}\right) \times \left(g_{(i)}(a) - g_{(i-1)}(a)\right)\right) = Ch(a).$$

Observe that the formulation of the area $\rho(S^*(a))$ expressed in terms of $\mu\left(\{g_j \in \mathcal{G} : g_j(a) \geq g_{(i)}(a)\}\right)$ can be used also when the capacity μ is not additive because there is some interaction between the criteria from $\mathcal{G}$. In other words, even if there are no weights $w_i,\ w_i \geq 0,\ i = 1, \dots, n,\ w_i + \dots + w_n = 1$, such that for $\mathcal{H} \subseteq \mathcal{G}$, $\mu(\mathcal{H}) = \sum_{g_i \in \mathcal{H}} w_i$, we can compute

$$\rho\left(R^*(a,(i))\right) = \mu\left(\{g_j \in \mathcal{G} : g_j(a) \geq g_{(i)}(a)\}\right) \times \left(g_{(i)}(a) - g_{(i-1)}(a)\right) = Ch(a).$$

The fact that $Ch(a)$ can be explained in terms of the area $\rho(S^*(a))$ explains why it is interpreted as an integral.

The Choquet integral can be expressed in a linear form (in a space different from the one of values taken by criteria from set $\mathcal{G}$) through the Möbius representation of the capacity μ as function $m : 2^{\mathcal{G}} \to \mathbb{R}$ (Rota, 1964; Shafer, 1976) such that, for all $\mathcal{H} \subseteq \mathcal{G}$

$$\mu(\mathcal{H}) = \sum_{\mathcal{K} \subseteq \mathcal{H}} m(\mathcal{K}).$$

In fact, Gilboa and Schmeidler (1994) proved that

$$Ch(a) = \sum_{\mathcal{H} \subseteq \mathcal{G}} m(\mathcal{H}) \min_{g_i \in \mathcal{H}} g_i(a). \tag{B.2}$$

To get a more flexible value function than the additive one, it is often sufficient to assume that interactions exist only between pairs of criteria, i.e., $m(\mathcal{H}) = 0$ for all $\mathcal{H} \subseteq \mathcal{G}$ such that $card(\mathcal{H}) > 2$ (Grabisch, 1997). In this case, the Choquet integral boils down to:

$$Ch(a) = \sum_{i=1}^{n} m_i g_i(a) + \sum_{g_{i_1}, g_{i_2} \in \mathcal{G},\ i_1 < i_2} m_{i_1 i_2} min(g_{i_1}(a), g_{i_2}(a)) \tag{B.3}$$

with $m_i = m(\{g_i\}) = \mu(\{g_i\})$ and $m_{i_1 i_2} = m(\{g_{i_1}, g_{i_2}\}) = \mu(\{g_{i_1}, g_{i_2}\}) - \mu(\{g_{i_1}\}) - \mu(\{g_{i_2}\})$. To illustrate this case, let us consider a decision problem involving three criteria g_1, g_2 and g_3, and capacity μ such that:

- $\mu(\emptyset) = 0,\ \mu(\{g_1, g_2, g_3\}) = 1,$
- $\mu(\{g_1\}) = 0.2,\ \mu(\{g_2\}) = 0.3,\ \mu(\{g_3\}) = 0.4,$
- $\mu(\{g_1, g_2\}) = 0.4,\ \mu(\{g_1, g_3\}) = 0.6,\ \mu(\{g_2, g_3\}) = 0.8.$

The Möbius representation of capacity μ is the following:

- $m(\emptyset) = 0,\ m(\{g_1, g_2, g_3\}) = 1,$
- $m(\{g_1\}) = 0.2,\ m(\{g_2\}) = 0.3,\ m(\{g_3\}) = 0.4,$
- $m(\{g_1, g_2\}) = -0.1,\ m(\{g_1, g_3\}) = 0,\ m(\{g_2, g_3\}) = 0.1.$

Supposing that for $a \in \mathcal{A},\ g_3(a) \leq g_1(a) \leq g_2(a)$, taking into account (B.1), the value of a in terms of the Choquet integral with respect to capacity μ is given by

$$\begin{aligned} Ch(a) =& g_3(a) \times \mu(\{g_j \in \mathcal{G} : g_j(a) \geq g_3(a)\}) + \\ & (g_1(a) - g_3(a)) \times \mu(\{g_j \in \mathcal{G} : g_j(a) \geq g_1(a)\}) + \\ & (g_2(a) - g_1(a)) \times \mu(\{g_j \in \mathcal{G} : g_j(a) \geq g_2(a)\}) = \\ & g_3(a) \times \mu(\{g_1, g_2, g_3\}) + (g_1(a) - g_3(a)) \times \mu(\{g_1, g_2\}) \\ & + (g_2(a) - g_1(a)) \times \mu(\{g_2\}) = \\ & g_3(a) \times 1 + (g_1(a) - g_3(a)) \times 0.4 + (g_2(a) - g_1(a)) \times 0.3. \end{aligned}$$

Using the Möbius representation, on the basis of (B.2), the Choquet integral can be written as follows:

$$\begin{aligned} Ch(a) =& g_1(a) \times m(\{g_1\} + g_2(a)) \times m(\{g_2\} + g_3(a)) \times m(\{g_3\}) + \\ & min\{g_1(a), g_2(a)\} \times m(\{g_1, g_2\}) + min\{g_1(a), g_3(a)\} \times m(\{g_1, g_3\}) + \\ & min\{g_2(a), g_3(a)\} \times m(\{g_2, g_3\}) + min\{g_1(a), g_2(a), g_3(a)\} \\ & \times m(\{g_1, g_2, g_3\}) = \\ & g_1(a) \times 0.2 + g_2(a) \times 0.3 + g_3(a) \times 0.4 + g_1(a) \times (-0.1) + \\ & g_3(a) \times 0 + g_3(a) \times 0.1 + g_3(a) \times 0.1. \end{aligned}$$

Consider now the capacity μ' such that $\mu'(\mathcal{H}) = \mu(\mathcal{H})$ for all $\mathcal{H} \subseteq \mathcal{G}$, with the exception of $\mathcal{H} = \{g_2, g_3\}$ for which $\mu'(\{g_2, g_3\}) = 0.9$. For the Möbius representation m' of μ', we have that for all $\mathcal{H} \subseteq \mathcal{G},\ m'(\mathcal{H}) = m(\mathcal{H})$, with the exception of $\mathcal{H} = \{g_2, g_3\}$ and $\mathcal{H} = \{g_1, g_2, g_3\}$ for which $m'(\{g_2, g_3\}) = 0.2$ and $m'(\{g_1, g_2, g_3\}) = 0$. Observe that, differently from μ, the capacity μ' is 2-additive because $m'(\mathcal{H}) = 0$ for all $\mathcal{H} \subseteq \mathcal{G}$ with $card(\mathcal{H}) \geq 2$ (in fact, the only subset $\mathcal{H}$ of $\mathcal{G}$ is $\{g_1, g_2, g_3\}$). As μ' is 2-additive, on the basis of (B.3), the Choquet integral can be expressed as follows:

$$\begin{aligned} Ch(a) =& m_1 \times g_1(a) + m_2 \times g_2(a) + m_3 \times g_3(a) + \\ & m_{1,2} \times min\{g_1(a), g_2(a)\} + m_{1,3} \times min\{g_1(a), g_3(a)\} \\ & + m_{2,3} \times min\{g_2(a), g_3(a)\} = \\ & 0.2 \times g_1(a) + 0.3 \times g_2(a) + 0.4 \times g_3(a) + (-0.1) \times g_1(a) \\ & + 0 \times g_3(a) + 0.5 \times g_3(a). \end{aligned}$$

It is worth noting that for specific values of capacities, the Choquet integral boils down not only to the weighted sum, but also to MIN and MAX, and OWA (Ordered Weighted Averaging) aggregation function (Yager, 1988) assigning each $a \in \mathcal{A}$ the value

$$OWA(a) = \sum_{i=1}^{n} w_i g_{(i)}(a), w_i \geqslant 0, i = 1, \dots, n, \sum_{i=1}^{n} w_i = 1.$$

Let us remember that another aggregation function based on the concept of capacity is the Sugeno integral (Sugeno, 1974). It has been designed for aggregation of criteria expressed on a common ordinal scale from 0 to 1. The Sugeno integral over evaluations of action $a \in \mathcal{A}$ on criteria $g_1, \dots, g_n$ is computed as follows:

$$Su\left(g_1(a), \dots, g_n(a)\right) = \max_{i=1,\dots,n} \left\{\min\left\{g_{(i)}(a), \mu\{g_j \in \mathcal{G} : g_j(a) \geq g_{(i)}(a)\}\right\}\right\} \tag{B.4}$$

with $g_{(1)}(a) \leq \dots \leq g_{(n)}(a)$ being the evaluations of action a on criteria from $\mathcal{G}$ ordered from the lowest to the highest.

Returning to the Choquet integral, it is worth mentioning two notable extensions: the bipolar Choquet integral (Grabisch & Labreuche, 2005) and the level dependent Choquet integral (Greco, Matarazzo, & Giove, 2011) (which has as special cases both the basic Choquet integral and the Sugeno integral).

An often criticized feature of the Choquet and Sugeno integrals is the requirement that the evaluations on all criteria are expressed on the same scale, which is hard to satisfy in real-world MCDA (Roy, 2009). The commensurability assumption has been abandoned in another popular nonlinear aggregation function able to handle interactions among criteria expressed on ratio scales – the multilinear model characterized by Keeney and Raiffa (1976) and Owen (1972), and recently

reconsidered by Pelegrina et al. (2020). Differently from the Choquet integral, the multilinear model is obtained as the linear interpolation using all vertices of the hypercube $[0;1]^n$ (Grabisch, 2016). The commensurability assumption has also been abandoned in the UTA^{GMS}-INT method (Greco et al., 2014) where instead of the Choquet integral a more general nonlinear aggregation function is used composed of an additive value function augmented by components modeling positive and negative interactions between criteria in pairs. This method implements the Robust Ordinal Regression methodology and is described in Section 7.5. Let us conclude this appendix by mentioning, among the various methods proposed to obtain the capacity needed to calculate the Choquet integral, the methodology proposed by Siskos and Burgherr (2022). This methodology is based on the disaggregation of the DM's Simos ranking of the criteria as well as other additional preferential data.

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